

Managerial Decision Making
Business 38802 (AXP 25)
Autumn 2025
George Wu

Class 3

Judgment under Uncertainty
(The Difficulty of Learning)

Course Outline

Monday, November 3	Introduction to Decision Making
Tuesday, November 4	Judgment under Uncertainty I (Heuristics and biases)
Wednesday, November 5*	Judgment under Uncertainty II (The difficulty of learning)
Friday, November 7	Risky Decision Making
Saturday, November 8	Nudges/Improving Decision Processes
January 5	Final (More later)

Sample Final Question

Consider the following fact: approximately 50% of small businesses fail within their first three years of operations. A friend is unaware of this fact and wishes to quit his or her job as an investment banker and open a restaurant in Barcelona. As his or her good friend, you want to be sure overconfidence is not playing a significant role in his or her decision. As you consider how to discuss this issue with your friend, consider the following questions:

- a. Why might your friend be overconfident about the success of this venture? Please be specific in identifying the most important sources of overconfidence.
- b. Which of these sources do you believe will be most resistant to your persuasion?

What did we learn yesterday?

- Two roles of judgment
 - Accuracy (~ best guess)
 - “Appropriate” confidence (~ confidence interval)
 - We are over-confident
 - We = AOL + You + Me
 - Why do we care?
 - Under-estimate risk
 - Fail to collect information
 - Fail to manage risk
 - Why?
 - Anchoring + availability
 - Poor feedback (Physicians vs. Weather forecasters)
 - More today...
-

Primary versus Secondary Knowledge

- Primary knowledge

- Trivia Pursuit test: “How much do you know?”

- Secondary knowledge

- Confucius test: “How much do you think you know?”

- “Shall I tell you what it is to know? To say you know what you know, and to say you do not know when you do not, that is knowledge.”

- Why study secondary knowledge?

- Make better decisions
 - Underestimate risk
 - Fail to collect information
 - Miss risk management opportunities

Primary versus Second Knowledge

■ Example 1

- ❑ Venture Capitalist investing in AI

■ Primary Knowledge

- ❑ Is this an innovative product?
- ❑ Are these leadership teams I want to invest in?
- ❑ ...
- ❑ More generally, what factors should I be thinking about and how do I learn about these factors?

■ Second Knowledge

- ❑ When is my expertise of limited value? (maybe product is just *too* novel, or the market is just *too* unfamiliar, etc.)
- ❑ More generally, what is outside of my control? The “Unknown unknowns”?

Primary versus Second Knowledge

- Example 2

- Hiring Manager

- Primary Knowledge

- What predicts success in my organization?

- Second Knowledge

- How much weight should I put on each of these factors?



How might poor second knowledge contribute to the "planning fallacy" (over-optimistic estimates for projects, such as HK HRL)?

Primary versus Second Knowledge

■ Example 3

- ❑ JC Penney hires Apple exec, Ron Johnson, to be CEO in 2011

■ Rationale for hiring

- ❑ Apple success: Johnson had developed Apple retail model (e.g., Apple Store and Genius Bar)

■ Rationale for firing

- ❑ JC Penney failure: Apple strategy did not work with customers used to sales. They did not want “fun places to hang out.”



How does this example illustrate questionable secondary knowledge for JC Penney board?

Gawande again...translated



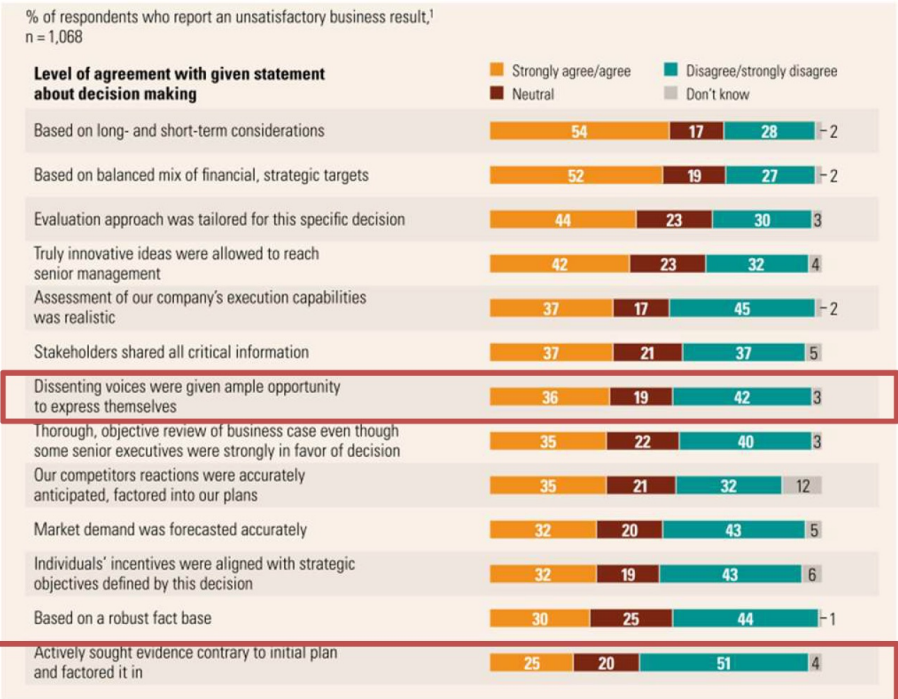
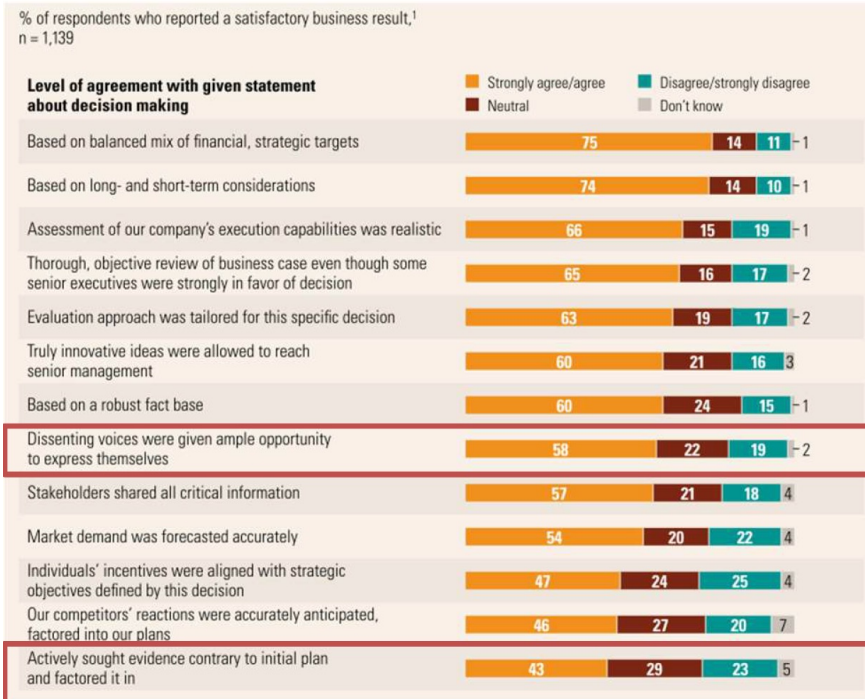
“The core predicament of business – the thing that makes being a decision maker so wrenching, being an executive so difficult, and being a part of a society that suffers the consequences of poor decision making – is uncertainty ... Business’ ground state is uncertainty. And wisdom – for managers of all ranks – is defined by how one copes with it.”

- Atul Gawande (2002), *Complications* (“The case of the red leg”)

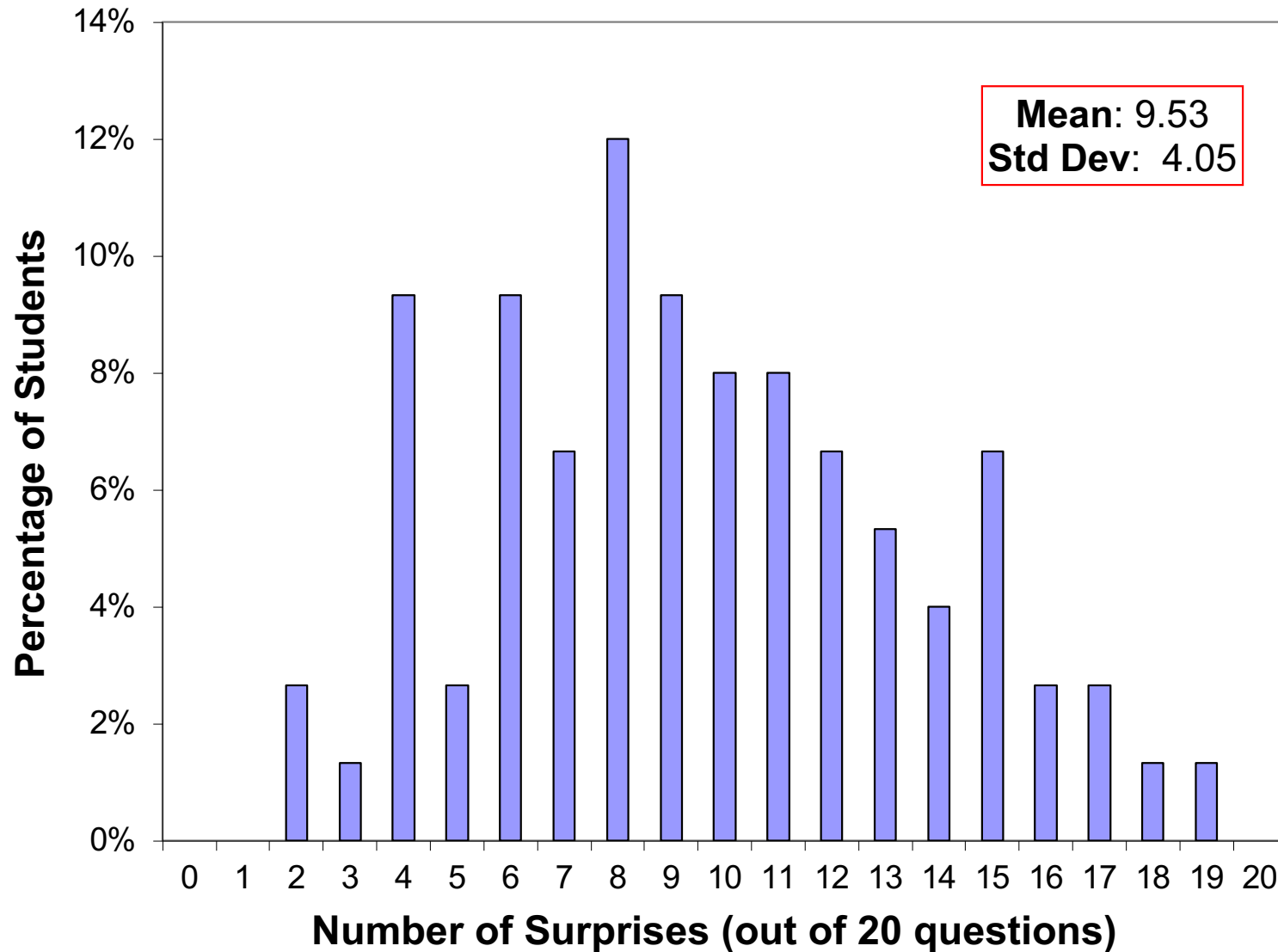
What did we learn yesterday?

■ Prescriptions

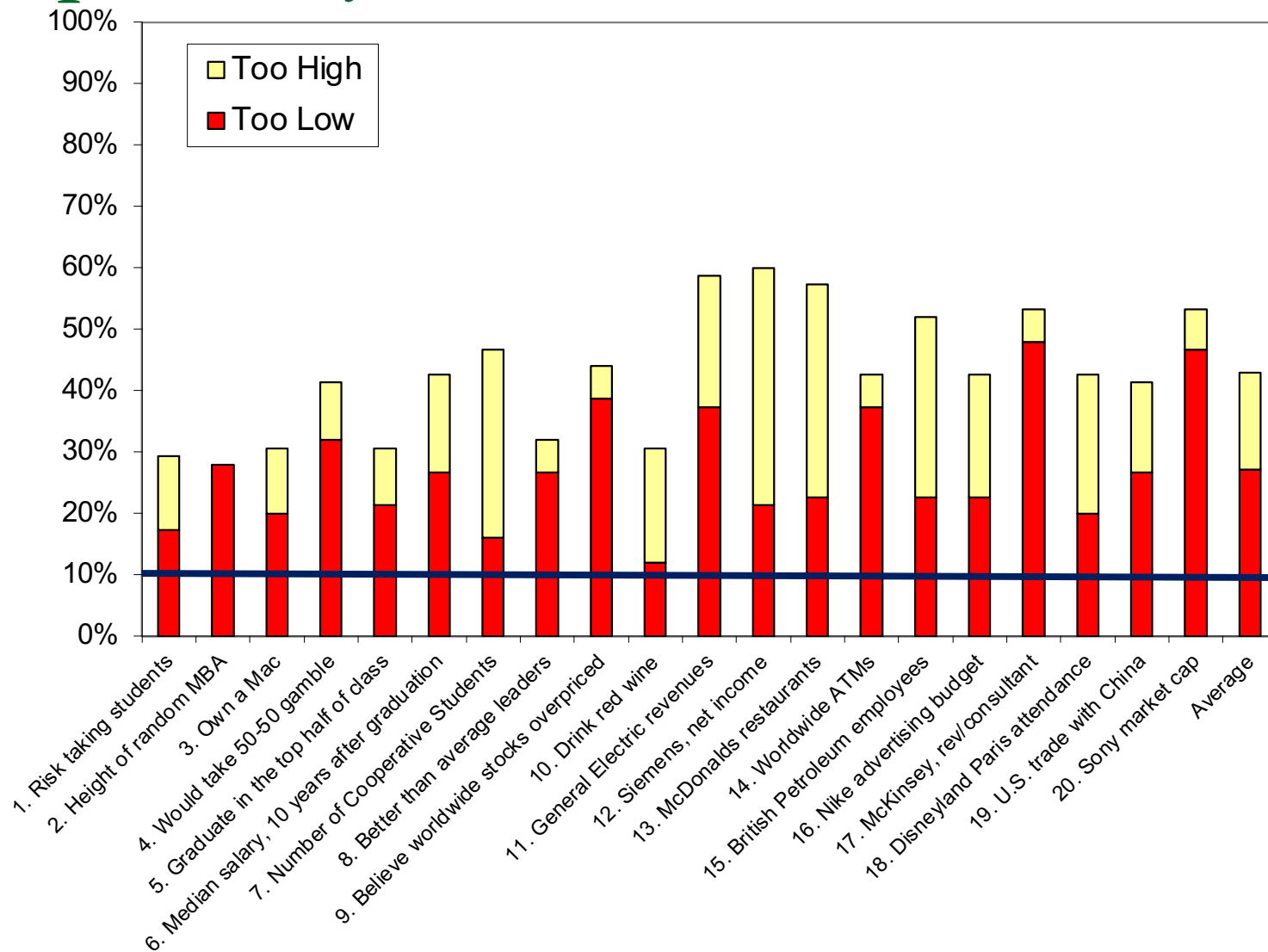
- Feedback
- Being contrarian
- More today



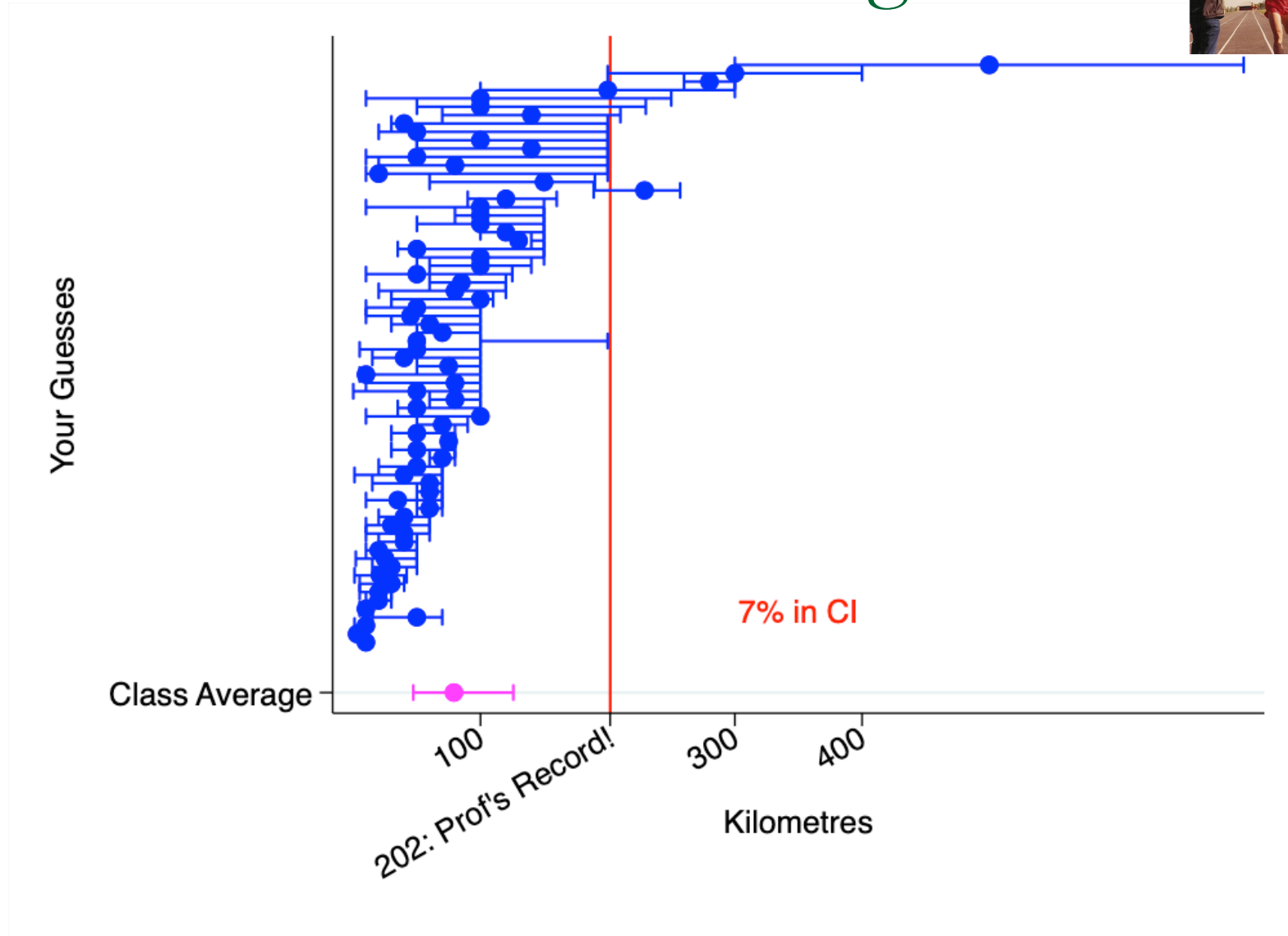
Surprises by Student



Surprises by Question



Professor Wu and Running



Some more wisdom...

■ Prescriptions

- ❑ Use multiple anchors (e.g., reference class forecasting or Bayes Rule)
- ❑ ~~Think in terms of confidence intervals~~
- ❑ Think in terms of better (wider?) confidence intervals
 - Use "Wisdom of crowds"
 - Take worse of worst (next and more in Class 5)

The “Wisdom of Crowds”

As he walked through the exhibition that day, Galton came across a weight-judging competition. A fat ox had been selected and placed on display, and members of a gathering crowd were lining up to place wagers on the weight of the ox. (Or rather, they were placing wagers on what the weight of the ox would be *after* it had been “slaughtered and dressed.”) For sixpence, you could buy a stamped and numbered ticket, where you filled in your name, your address, and your estimate. The best guesses would receive prizes.

Eight hundred people tried their luck. They were a diverse lot. Many of them were butchers and farmers, who were presumably expert at judging the weight of livestock, but there were also quite a few people who had, as it were, no insider knowledge of cattle. “Many non-experts competed,” Galton wrote later in the scientific journal *Nature*, “like those clerks and others who have no expert knowledge of horses, but who bet on races, guided by newspapers, friends, and their own fancies.” The analogy to a democracy, in which people of radically different abilities and interests each get one vote, had suggested itself to Galton immediately. “The average competitor was probably as well fitted for making a just estimate of the dressed weight of the ox, as an average voter is of judging the merits of most political issues on which he votes,” he wrote.

A NEW YORK TIMES BUSINESS BESTSELLER

“As entertaining and thought-provoking as *The Tipping Point* by Malcolm Gladwell. . . . *The Wisdom of Crowds* ranges far and wide.”
—*The Boston Globe*

THE WISDOM OF CROWDS

JAMES
SUROWIECKI

WITH A NEW AFTERWORD BY THE AUTHOR



The “Wisdom of Crowds”

THE WISDOM OF CROWDS

tests on them. Galton arranged the guesses (which totaled 787 in all, after he had to discard thirteen because they were illegible) in order from highest to lowest and graphed them to see if they would form a bell curve. Then, among other things, he added all the contestants’ estimates, and calculated the mean of the group’s guesses. That number represented, you could say, the collective wisdom of the Plymouth crowd. If the crowd were a single person, that was how much it would have guessed the ox weighed.

Galton undoubtedly thought that the average guess of the group would be way off the mark. After all, mix a few very smart people with some mediocre people and a lot of dumb people, and it seems likely you’d end up with a dumb answer. But Galton was wrong. The crowd had guessed that the ox, after it had been slaughtered and dressed, would weigh 1,197 pounds. After it had been slaughtered and dressed, the ox weighed 1,198 pounds. In other words, the crowd’s judgment was essentially perfect. Perhaps breeding did not mean so much after all. Galton wrote later: “The result seems more creditable to the trustworthiness of a democratic judgment than might have been expected.” That was, to say the least, an understatement.

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THE WISDOM OF CROWDS

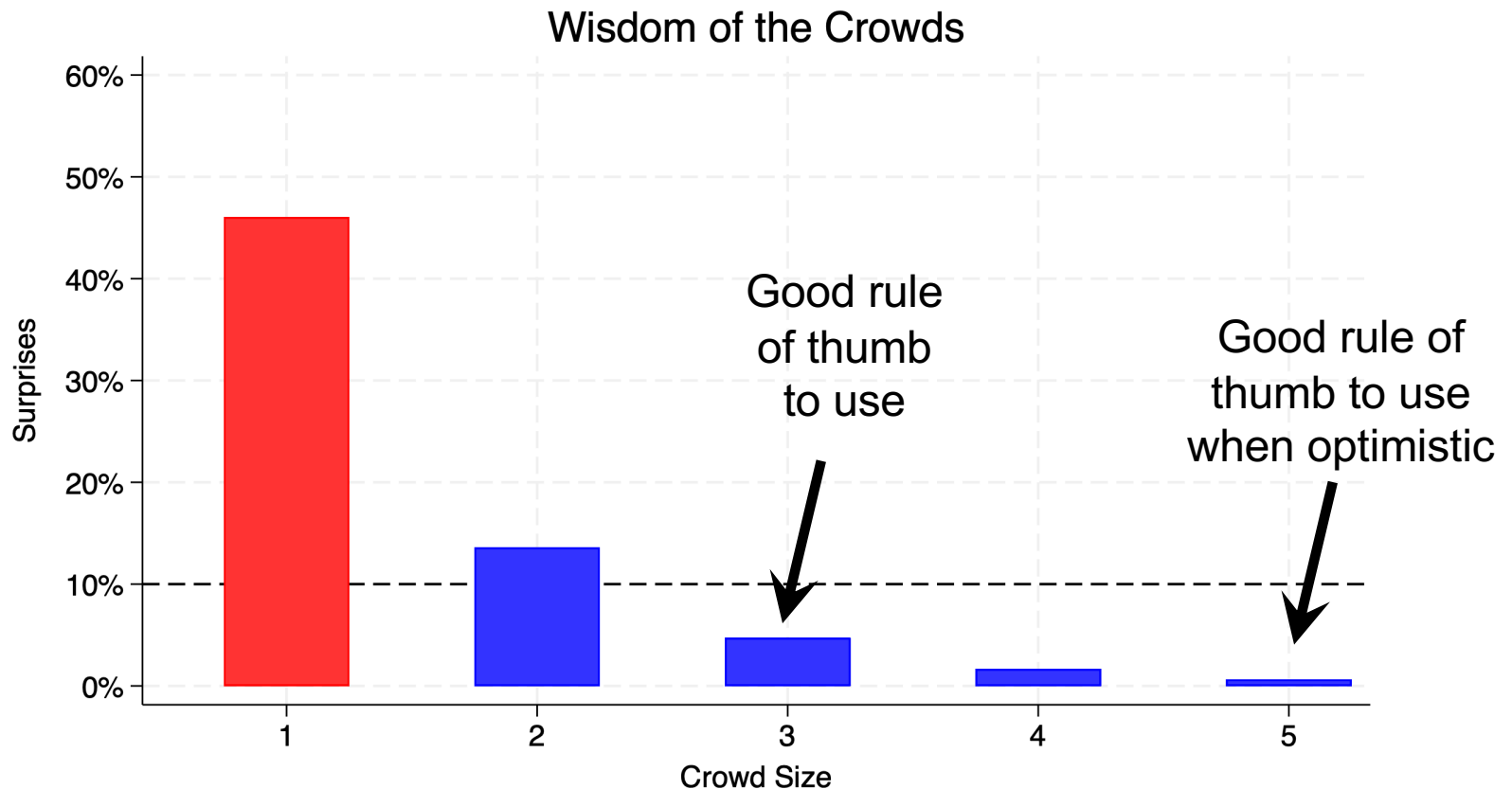
JAMES
SUROWIECKI

WITH A NEW AFTERWORD BY THE AUTHOR



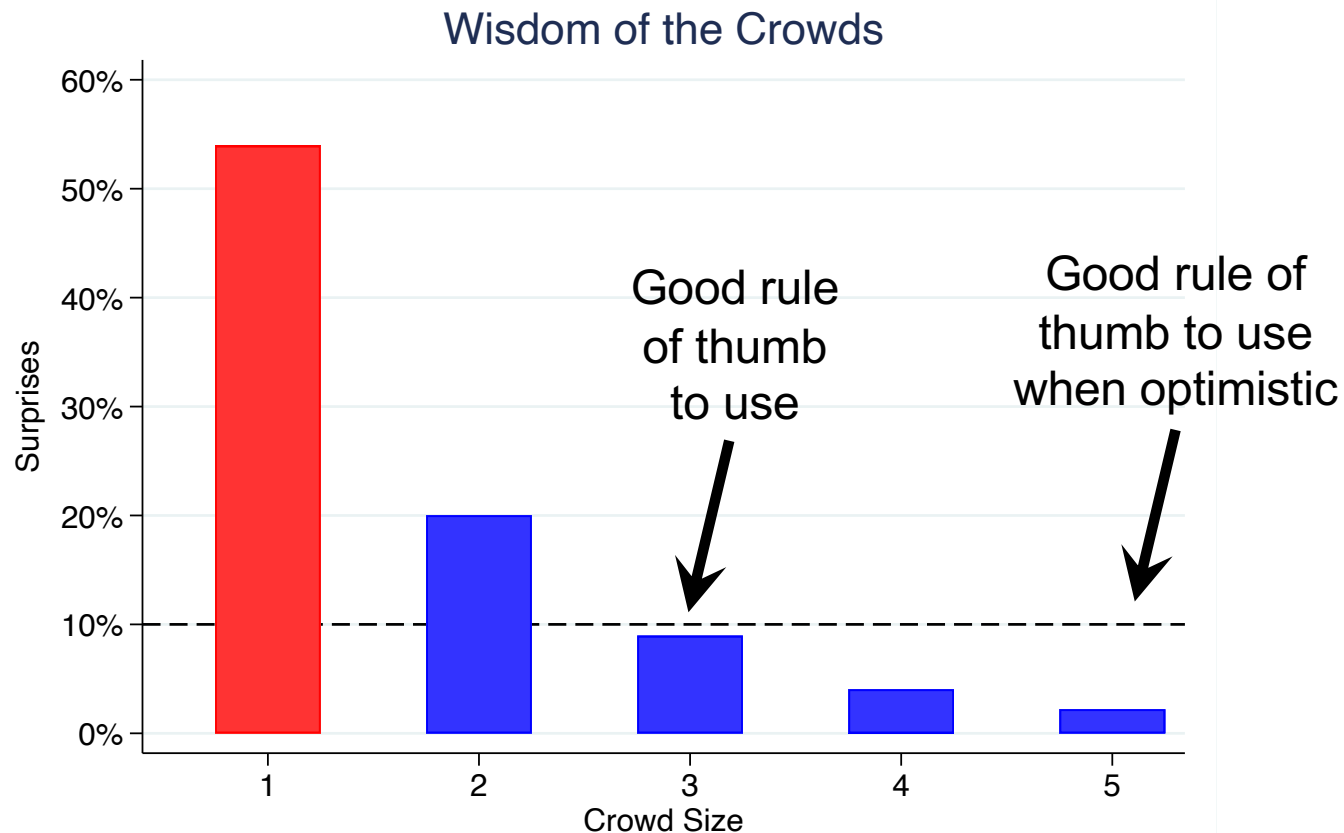
Using the “Wisdom of Crowds”

- Low (.05 fractile) is **lowest** of group's low (“min min”)
- High (.95 fractile) is **highest** of group's high (“max max”)



Using the “Wisdom of Crowds” (2024)

- Low (.05 fractile) is **lowest** of group’s low (“min min”)
- High (.95 fractile) is **highest** of group’s high (“max max”)



Today's Objectives

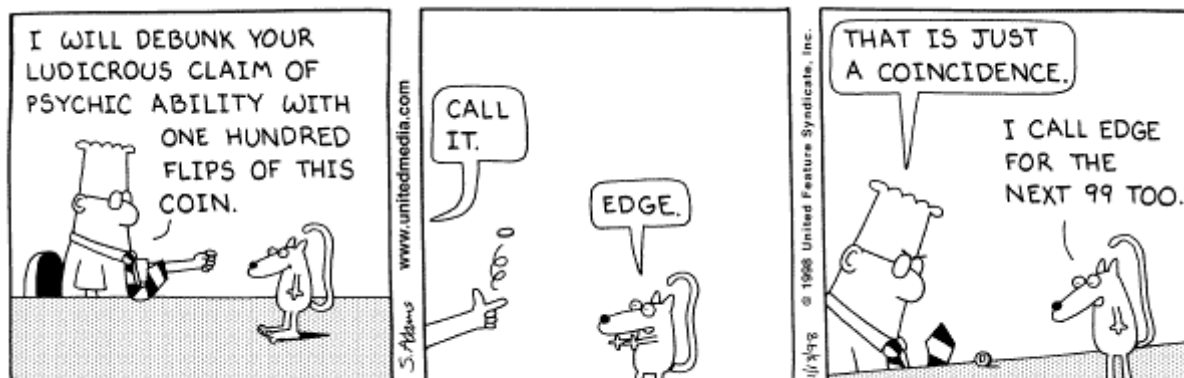
- Goals: The Difficulty of Learning
 - Why feedback is necessary but not sufficient
 - Two causes: Confirmation and hindsight biases
- Outline
 - First half: lecture sprinkled with examples
 - Second half: Discussion of *Into Thin Air*

Feedback is necessary, not sufficient

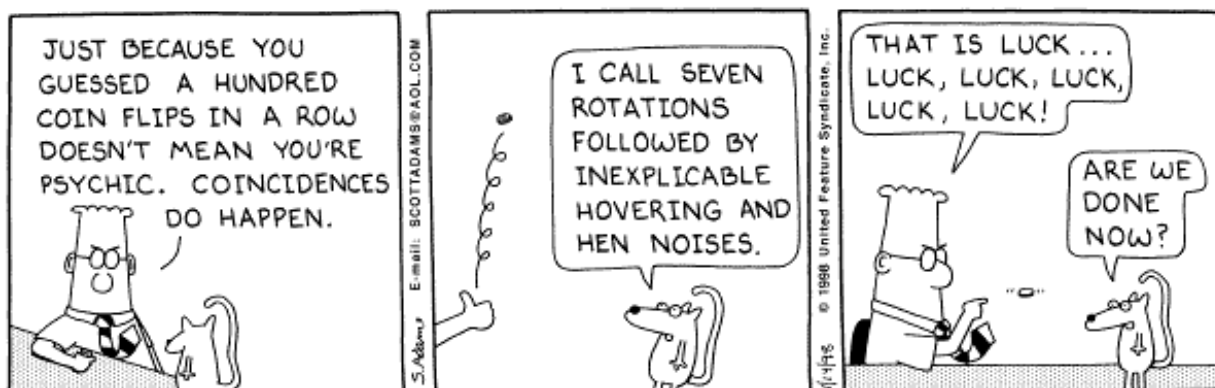
- Why?



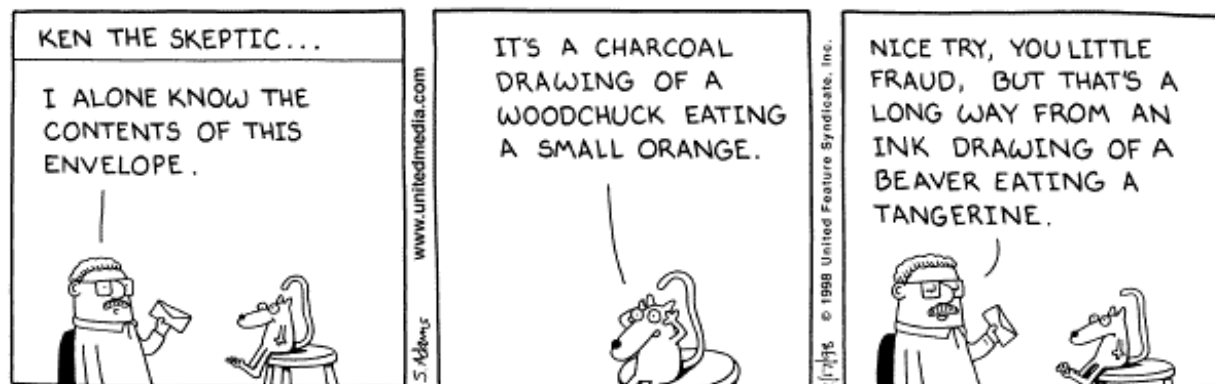
Think about a routine decision practice used in your organization. How would you determine if that practice was a good one? the best one?



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Algorithms vs. People

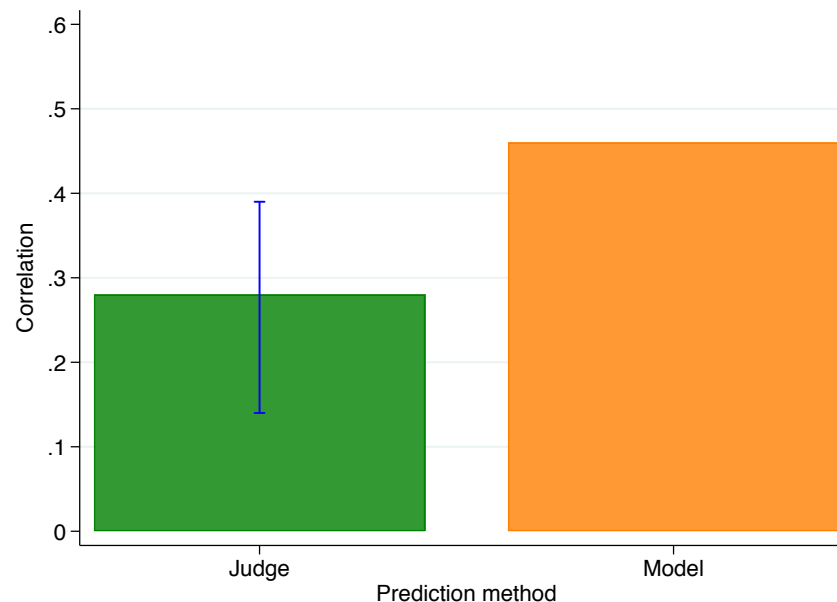
- Linear Models (now, AI and Machine Learning)
 - Used in credit selection, some college admissions; wine prediction; etc.
 - Goal: Predict Y based on $x_1, \dots, x_n$
 - Example
 - Y : American Express profitability if card issued to customer
 - x_i : salary, age, credit history, debt, mortgages, etc.
 - Two models
 - Statistical Model (Fit past data), regression/machine learning
 - Expert's Intuitive Judgment
 - Outcome measure: Correlation (r)
 - 0 = no correlation, no predictive ability
 - close to 0: weak predictors of actual outcome
 - close to 1 strong predictors of actual outcome
 - 1 = perfect correlation

Example 1: Clinical Psychologists

Judges: 29 clinical psychologists

Judgment: whether a patient is neurotic or psychotic (as later diagnosed by someone)

Data: 11 scores from Minnesota Multiphasic Personality Test

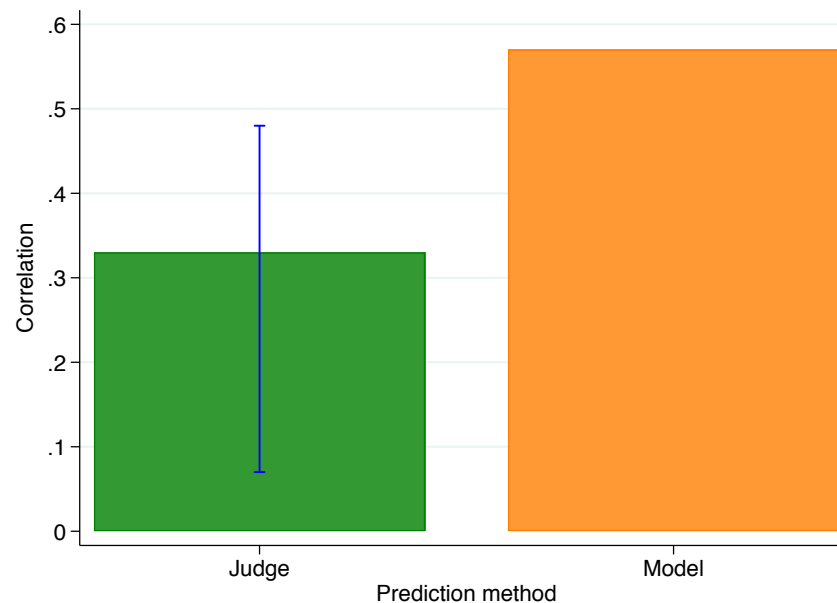


Example 2: Graduate Students

Judges: 80 graduate students

Judgment: predict first year grades for incoming students

Data: 10 predictor variables, including undergraduate GPA, GRE score, etc.

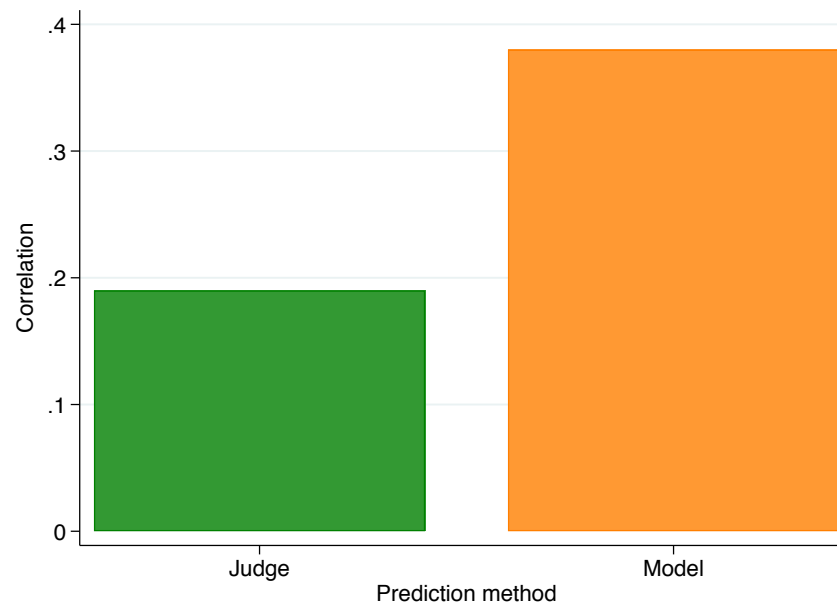


Example 3: Faculty

Judges: Admissions Committee

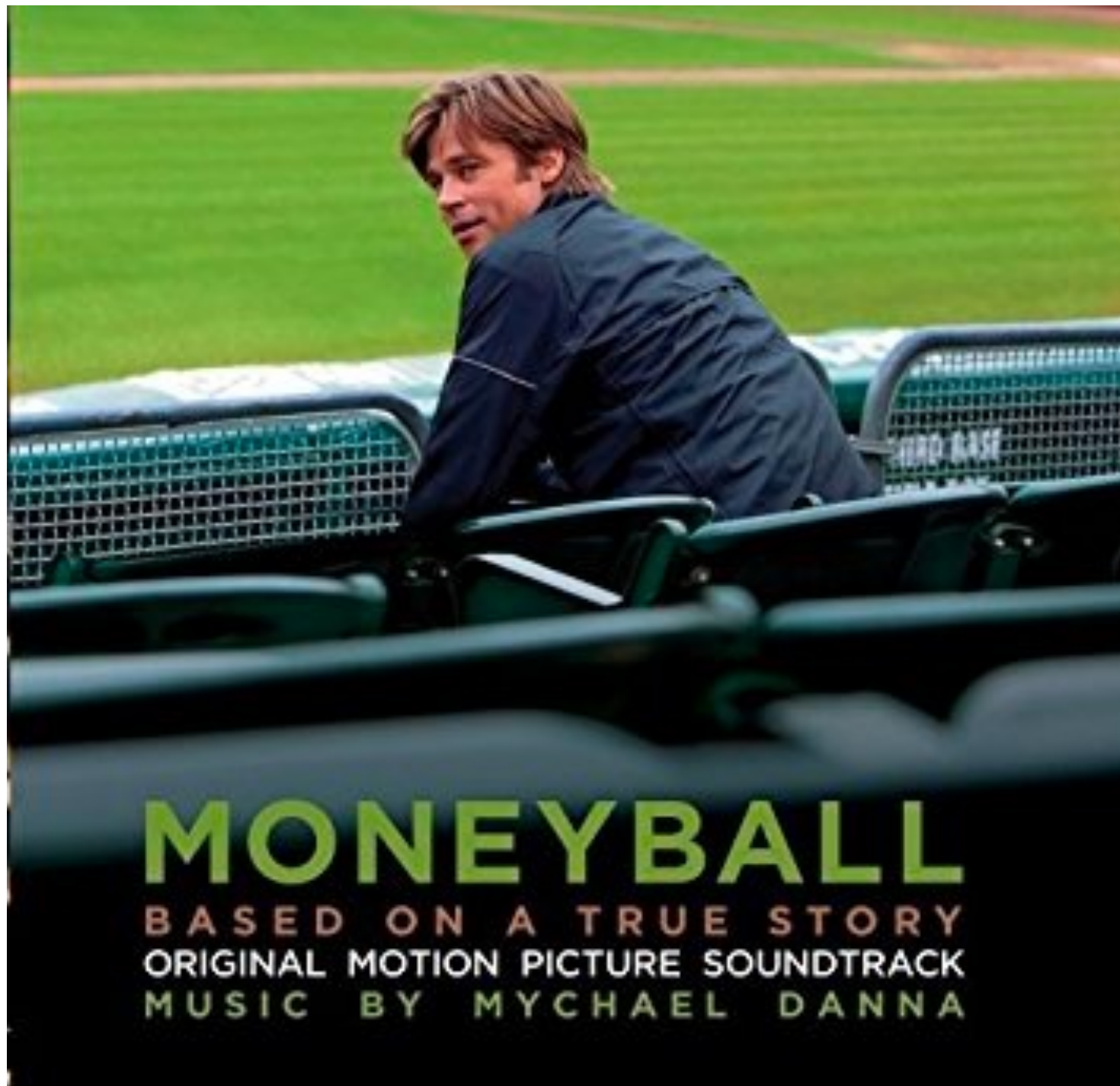
Judgment: predicting success in graduate school

Data: application, college grades, tests, etc.



Application: Interviews

- Interviews have surprisingly little predictive power
 - Why?



MONEYBALL

BASED ON A TRUE STORY
ORIGINAL MOTION PICTURE SOUNDTRACK
MUSIC BY MYCHAEL DANNA

Human Judgment vs. Algorithms

- Humans consider things they shouldn't
 - ❑ We create narratives and explanations (more later)
- Humans don't consider things they should
 - ❑ One narrative excludes other narratives
 - ❑ Some relevant information is not included
- Humans don't know how much to weigh each piece of evidence
 - ❑ What information is **not diagnostic** (thus, should get low weight)?
 - ❑ What information is **redundant** (thus, should also get low weight)?
- Humans are noisy, inconsistent, and influenced by
 - ❑ Fatigue
 - ❑ Recent judgments
 - ❑ Order of information

Human Judgment vs. Algorithms

- Algorithms consider the “right” pieces of evidence
- Algorithms weight the evidence appropriately and consistently
- Nevertheless, for consequential forecasts, people often choose to ignore algorithms
 - Eastwood, Snook, and Luther, 2012; Diab, Pui, Yankelovich, & Highhouse, 2011; Shaffer et al., 2013; Onkal et al., 2009; Promberger & Baron, 2006; Mahmoud et al, 2022
- Why?



Identify a situation in which you chose a person over an algorithm. Was this appropriate?

Why do people distrust algorithms?

Dietvorst, Simmons, and Massey (2015)

- People and algorithms both make mistakes
- We see errors in both types of “recommendations”, but
 - We are more tolerant of human errors
 - Errors made by algorithms are salient (i.e., available) (e.g., self-driving cars)
- We lose confidence in an algorithm after seeing it err

Why do people distrust algorithms?

Dietvorst, Simmons, and Massey (2015)

- (False?) Belief that Humans > Models at:
 - ❑ Use cues that aren't included in the model
 - ❑ Find complex patterns
 - ❑ Detect exceptions
 - ❑ Learn from mistakes
 - ❑ Get better with practice

Why do people distrust algorithms?

Sunstein and Gaffe (2024)

- (Inappropriately) high levels of confidence in human decision maker
- Individual agency and control
 - Self-navigation versus GPS
- Asymmetry between how we make sense of algorithm and human mistakes
- Low levels of comfort with innovating and technology
- Perceived suitability for algorithmic decision making
 - Imagine algorithm that gives name to your child

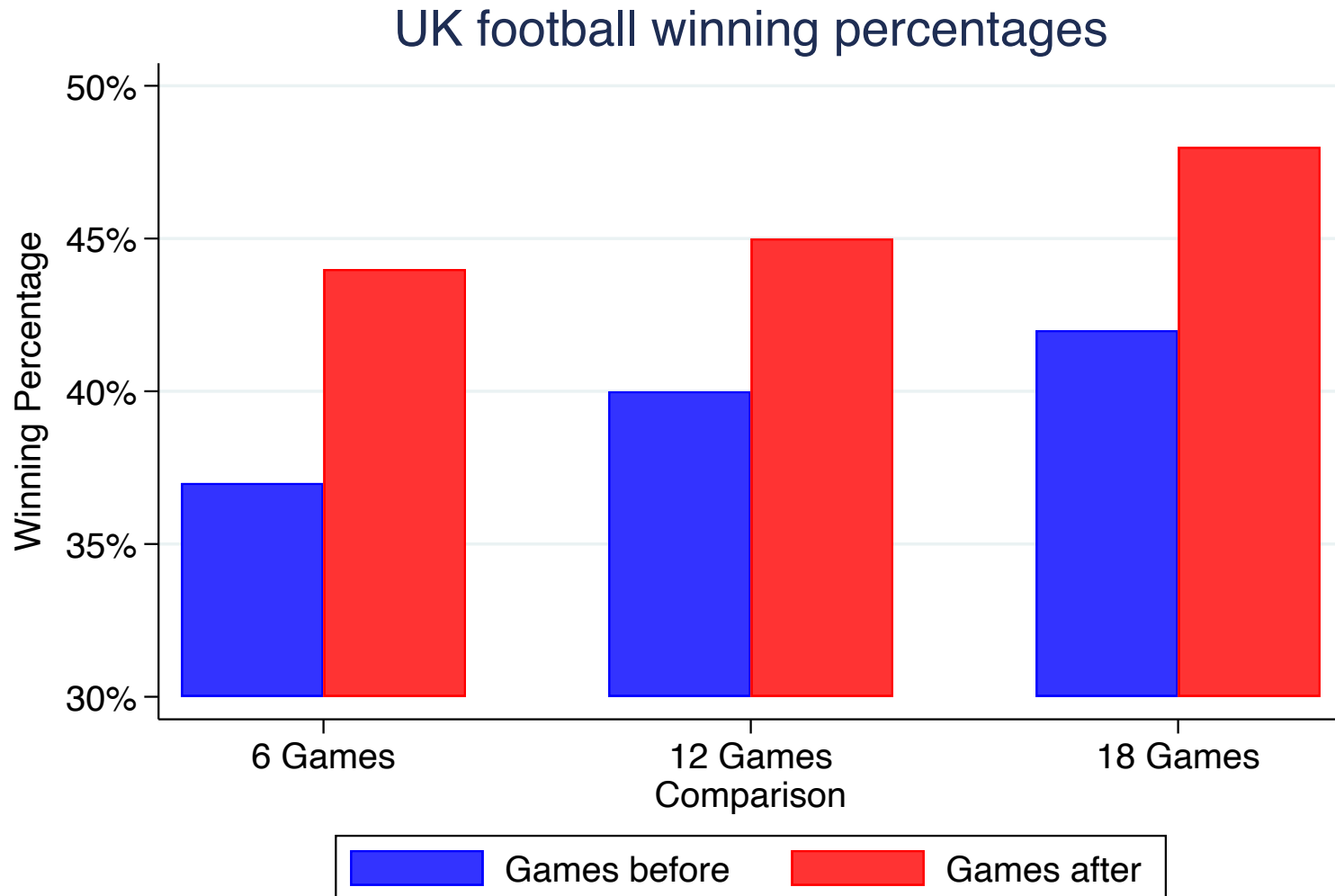
Managerial Turnover

- Belief that firing management is a good thing
 - When do you fire your management?
 - Usually when things aren't going well!
 - How do you prove that you're right?
 - Look at what happens after management turns over
 - Study: Examination of winning percentages of U.K. football teams
 - Empirical Question:
 - What happens when we fire our coach?
-

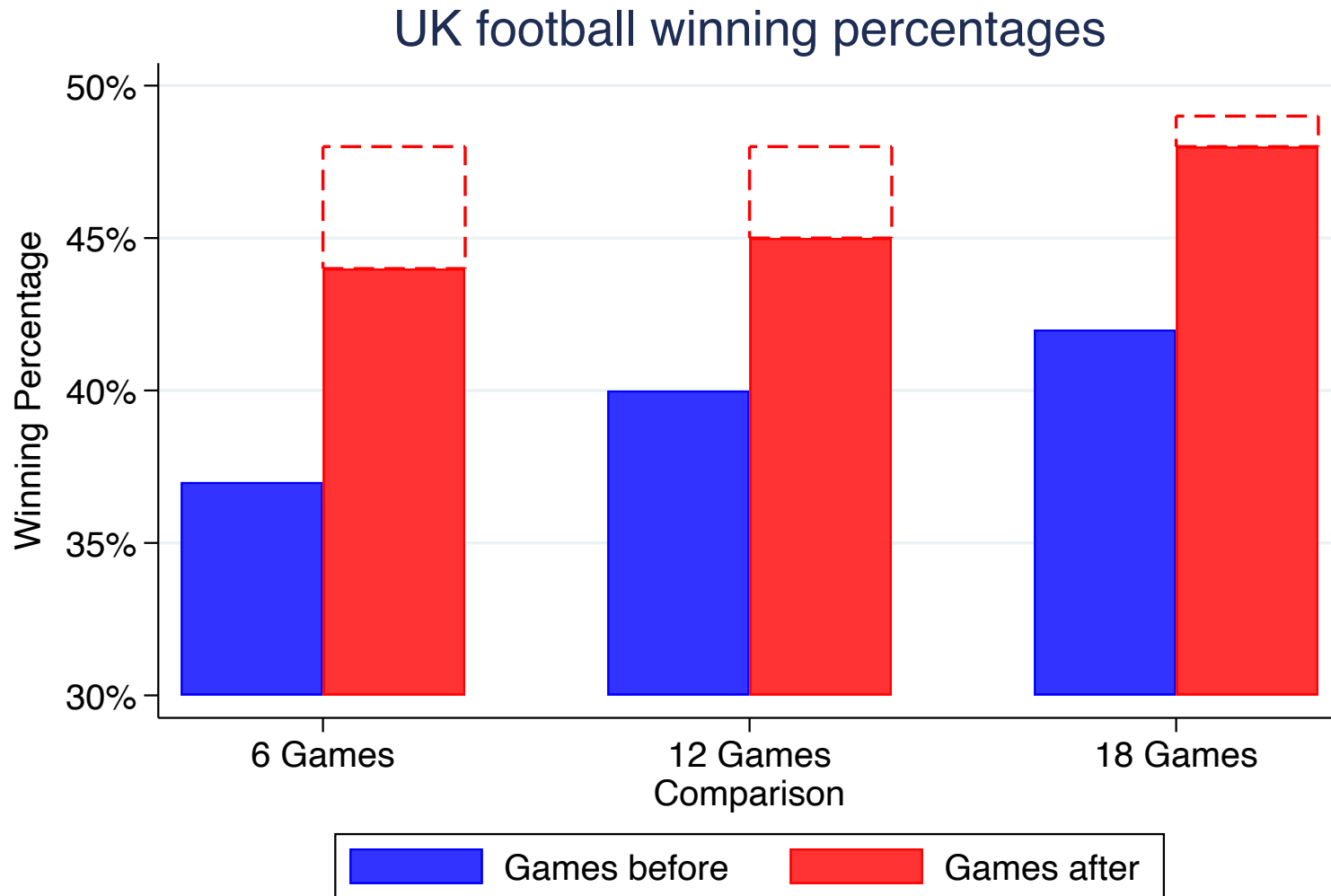
Managerial Turnover



Let's look at more data ...

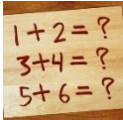


What about teams that didn't terminate?



Dotted area: incremental winning percentage for teams which did not fire manager

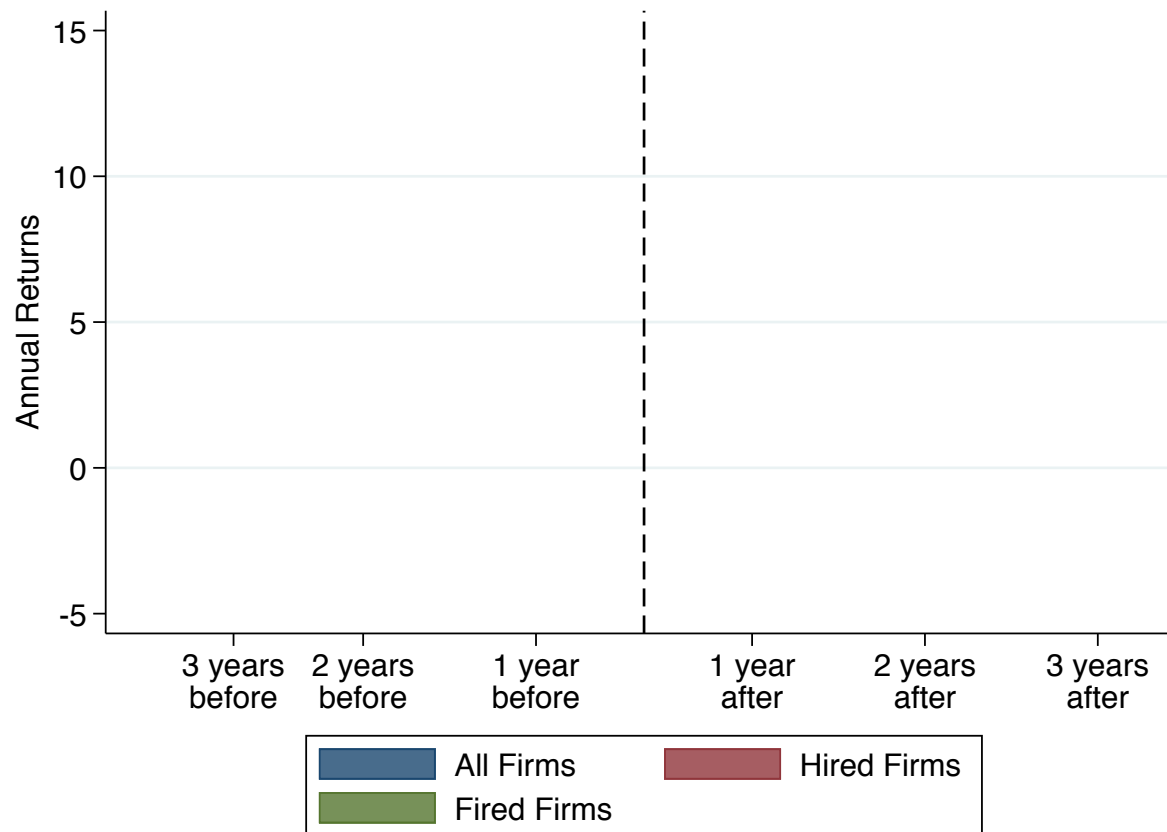
What about teams that didn't terminate?



Why does regression to the mean offer an alternative explanation for improvement after termination?

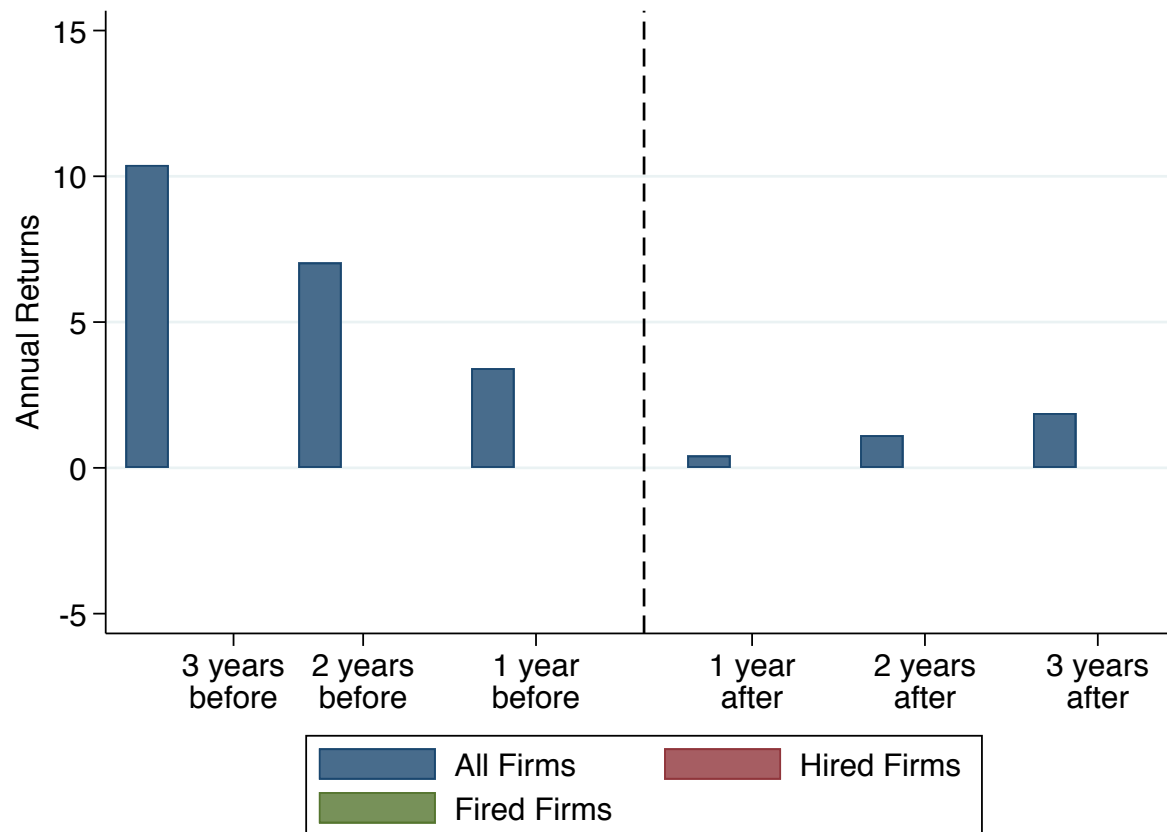
Managerial Turnover

- Study: Hiring and Firing of Investment Managers (Goyal and Wahal, 2008)



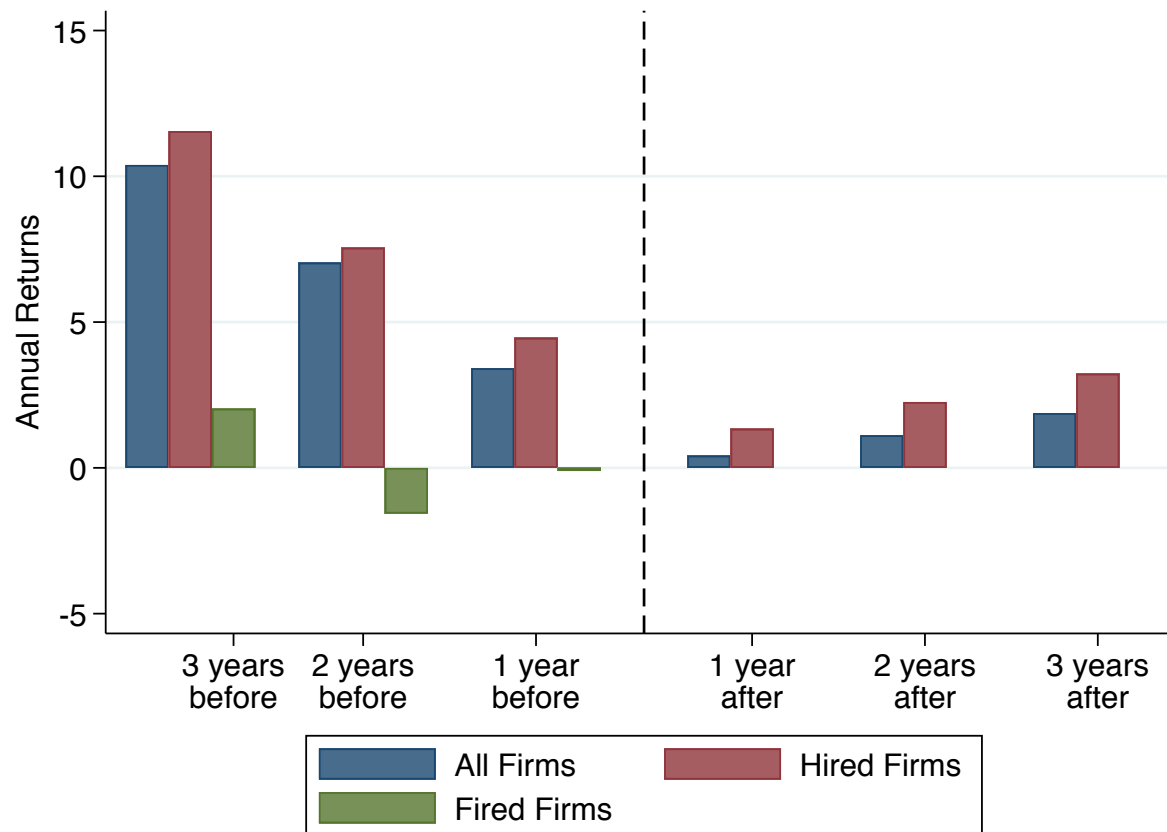
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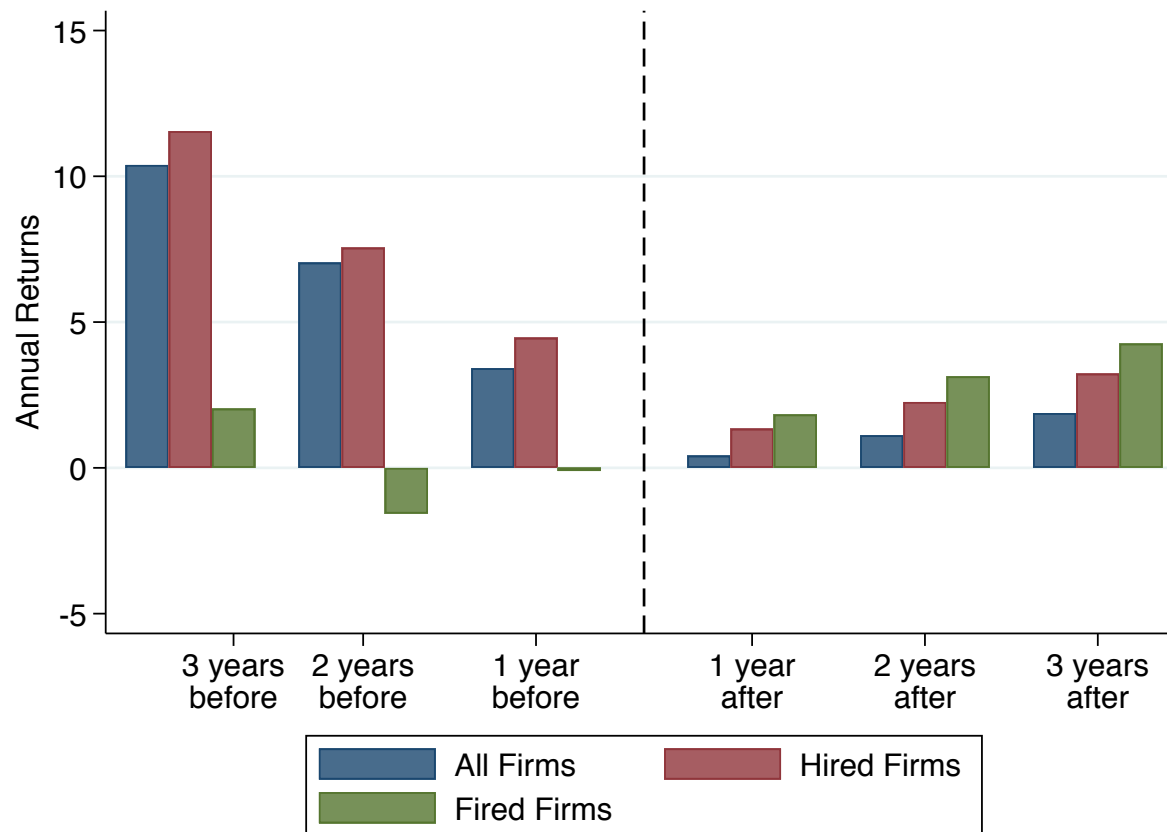
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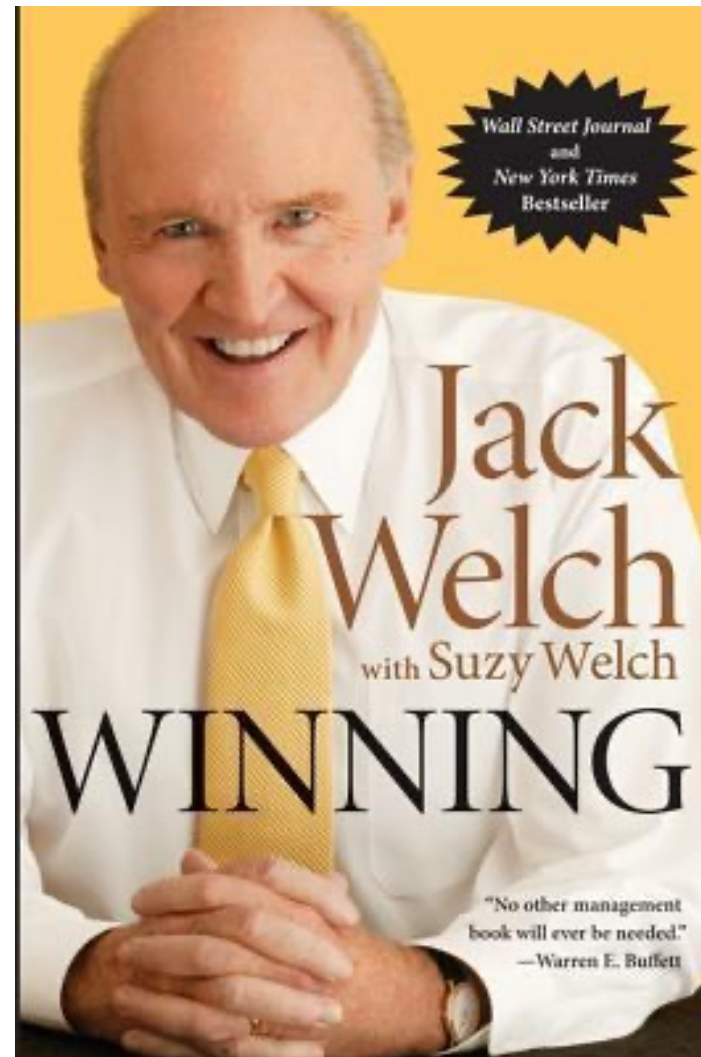
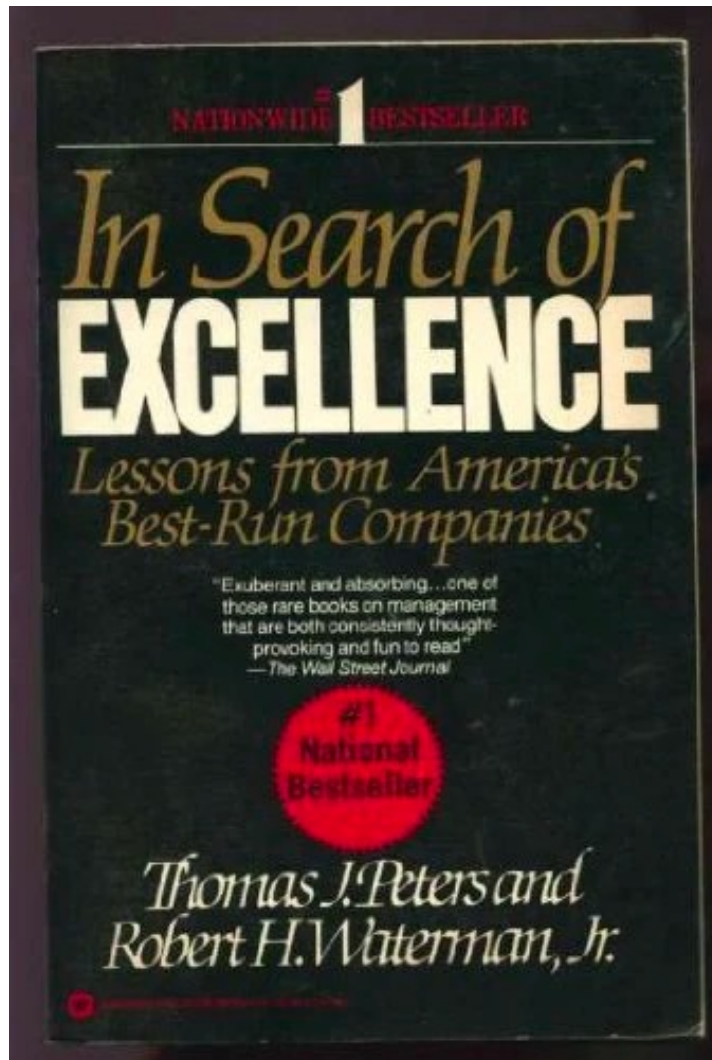


Managerial Turnover

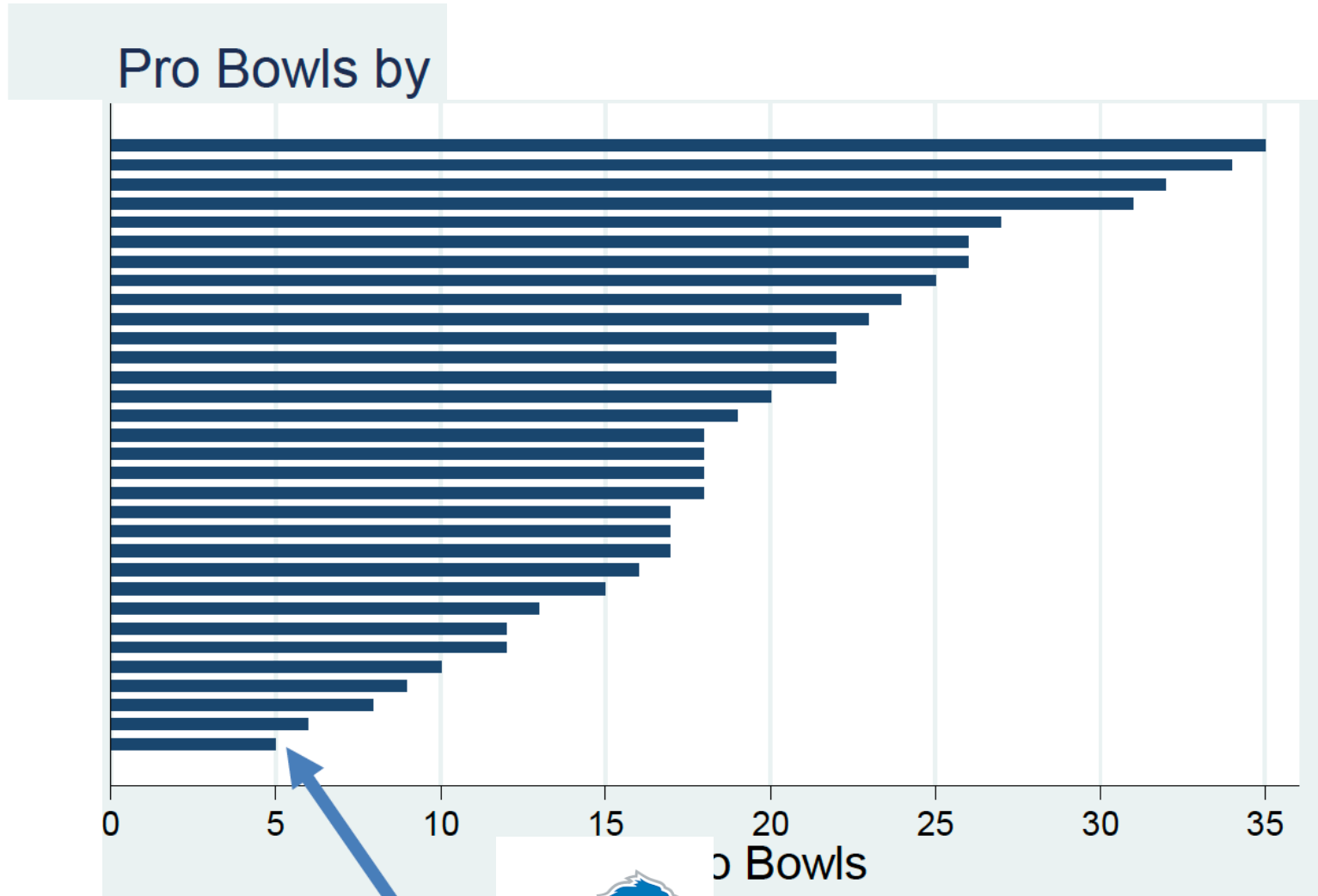
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Successful CEOs and Firms



Successful CEOs and Firms

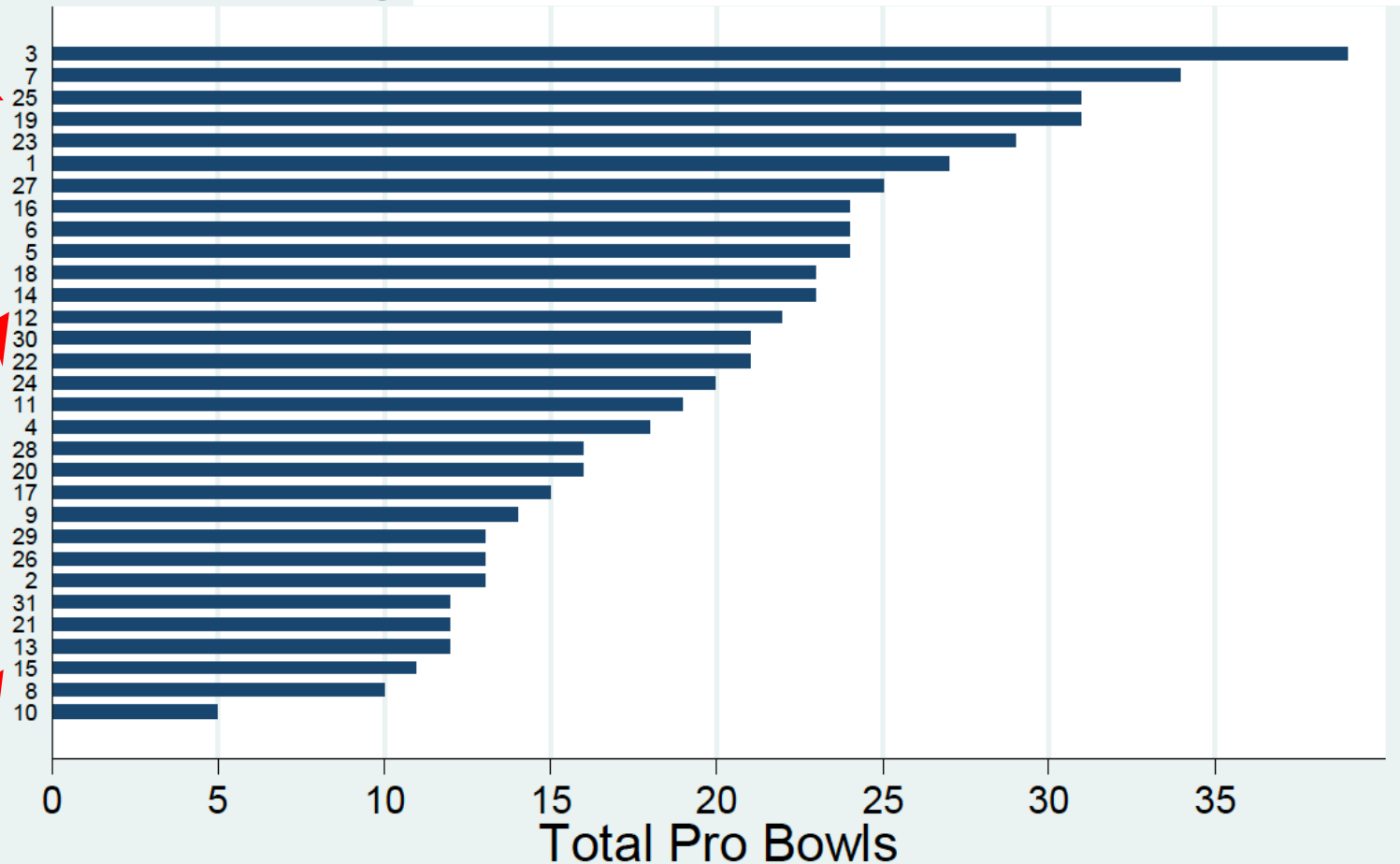


Successful CEOs and Firms

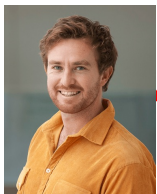
My day
of the
month! 😊



Pro Bowls by



1997-2007 drafts, through 2009 season.



Golf

- Belief that some golfers “choke”
 - ❑ When they are close to the lead
 - ❑ When they are playing in big tournaments
 - ❑ When they are playing “above their head”



Golf

2017 British Open: Justin Thomas (USA)



First Round (67-T6)

- “People began the day at the British Open talking about what Justin Thomas was wearing. They’ll end it talking about the way he played.”

Second Round (80-T minus 6)

- “After opening 67 wearing a necktie, he choked away his chances with a second-round 80 including a quintuple 9 to accelerate meltdown.”

Golf

2018 British Open: Jon Rahm (Spain)



First Round (69-T8)

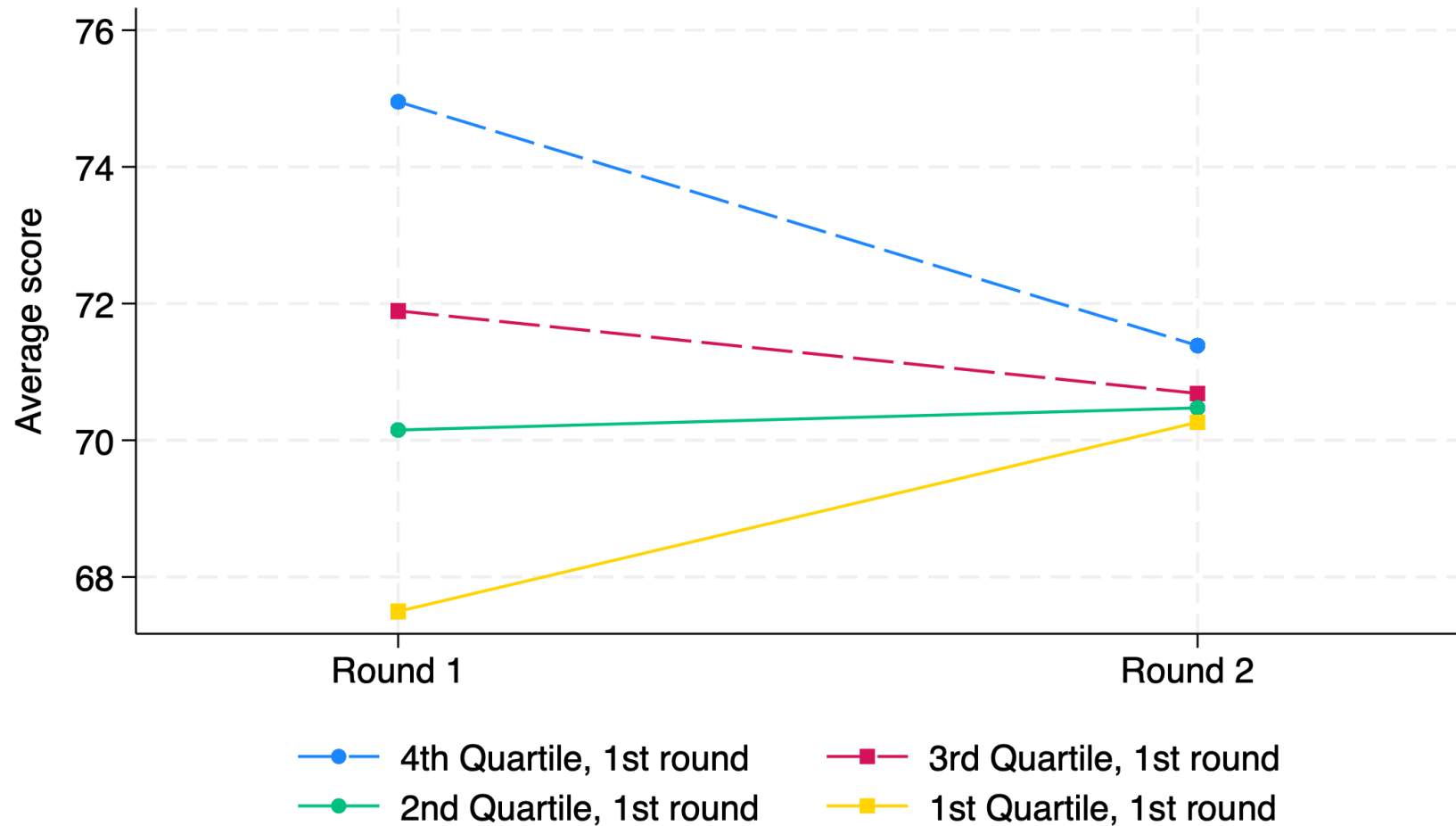
- “After Day 1 at Carnoustie, Rahm and McIlroy are listed as the new favorites for the claret jug, owning 10-1 odds by Ladbrokes, William Hill and Sky Bet.”

Second Round (78)

- “Jon Rahm loses his cool and his Major dream as he he drops six shots in three holes and throws his club.”
-

PGA Golf Scores: 1st vs. 2nd Rounds

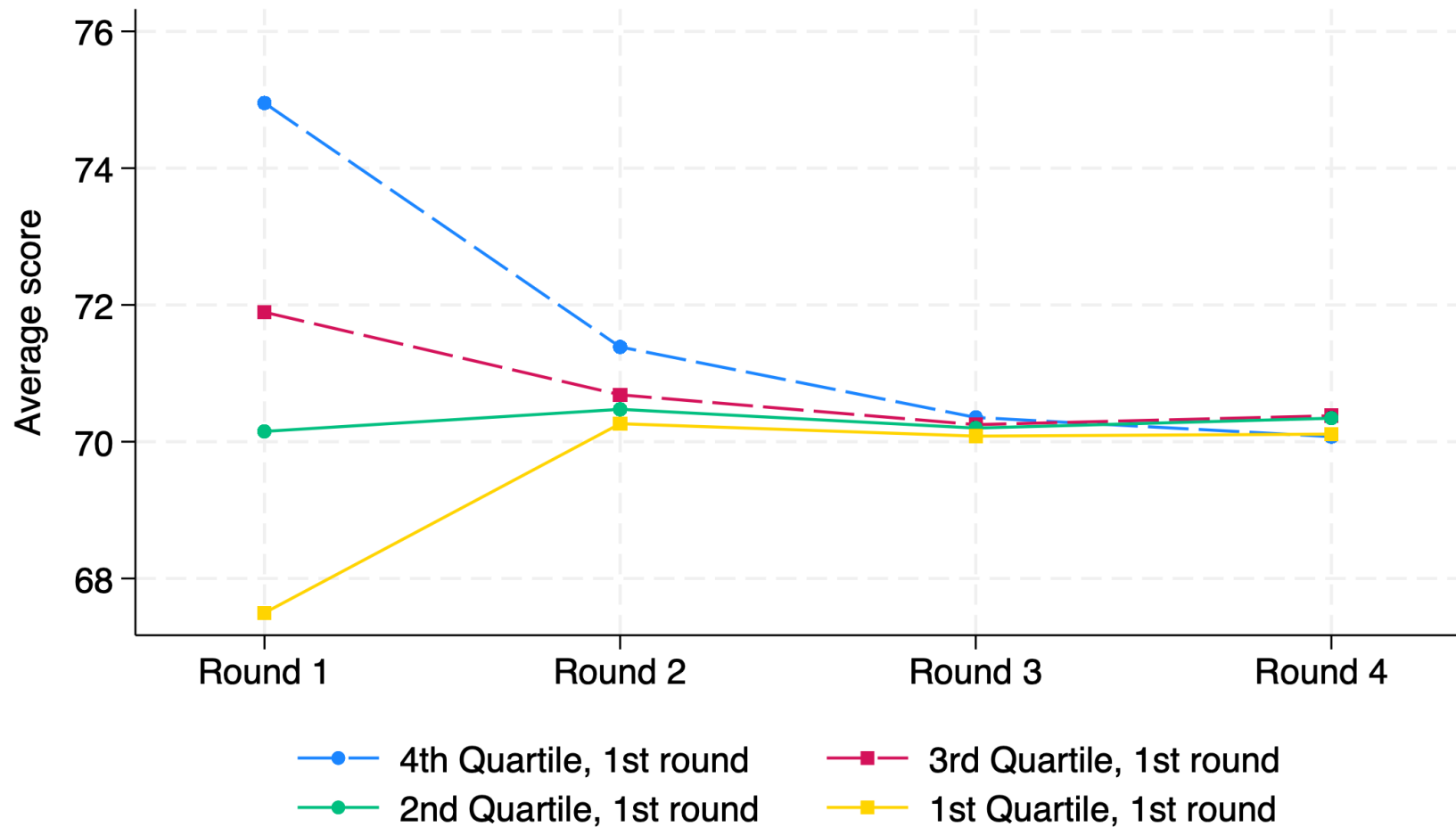
Persistence in Professional Golf



2001-2025 PGA tour (n = 262,418 golfer rounds)

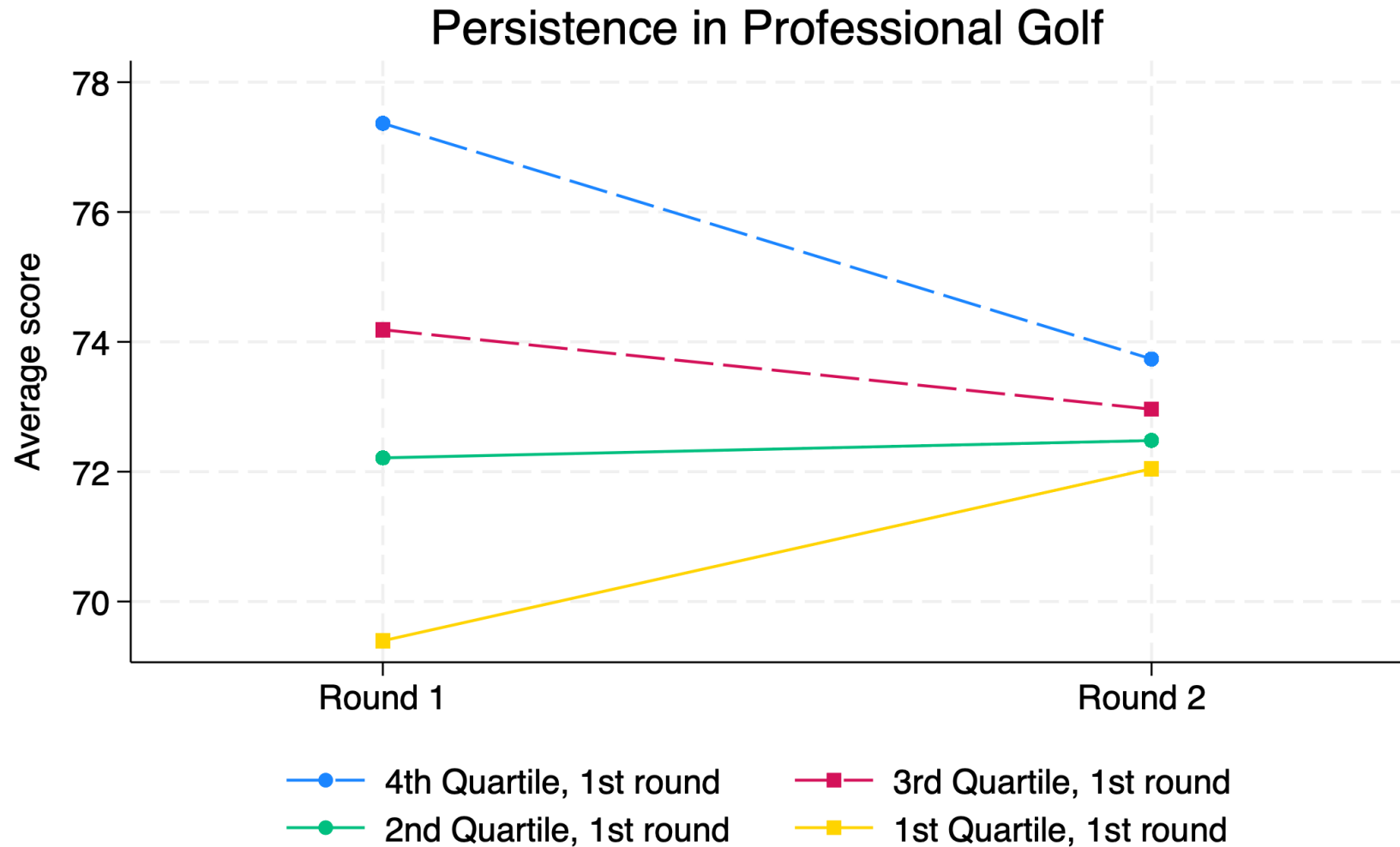
PGA Golf Scores: 1st vs. Other Rounds

Persistence in Professional Golf



2001-2025 PGA tour (n = 524,836 golfer rounds)

LPGA Golf Scores: 1st vs. 2nd Rounds



2001-2025 LPGA tour (n = 334,342 golfer rounds)

Signal vs. Noise in Golf

Scottie Scheffler

(1)



Tommy Fleetwood

(2)



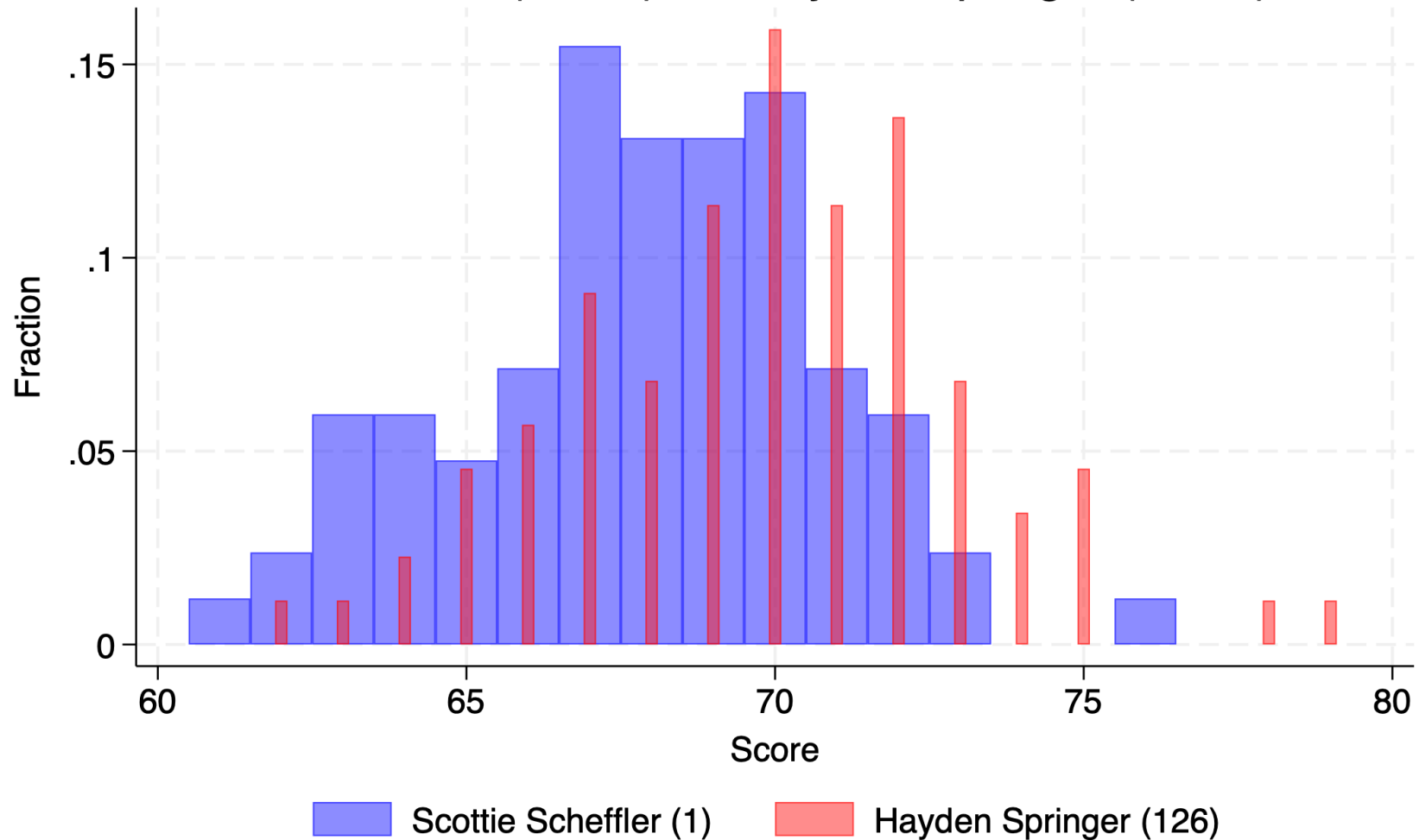
Hayden Springer

(126)



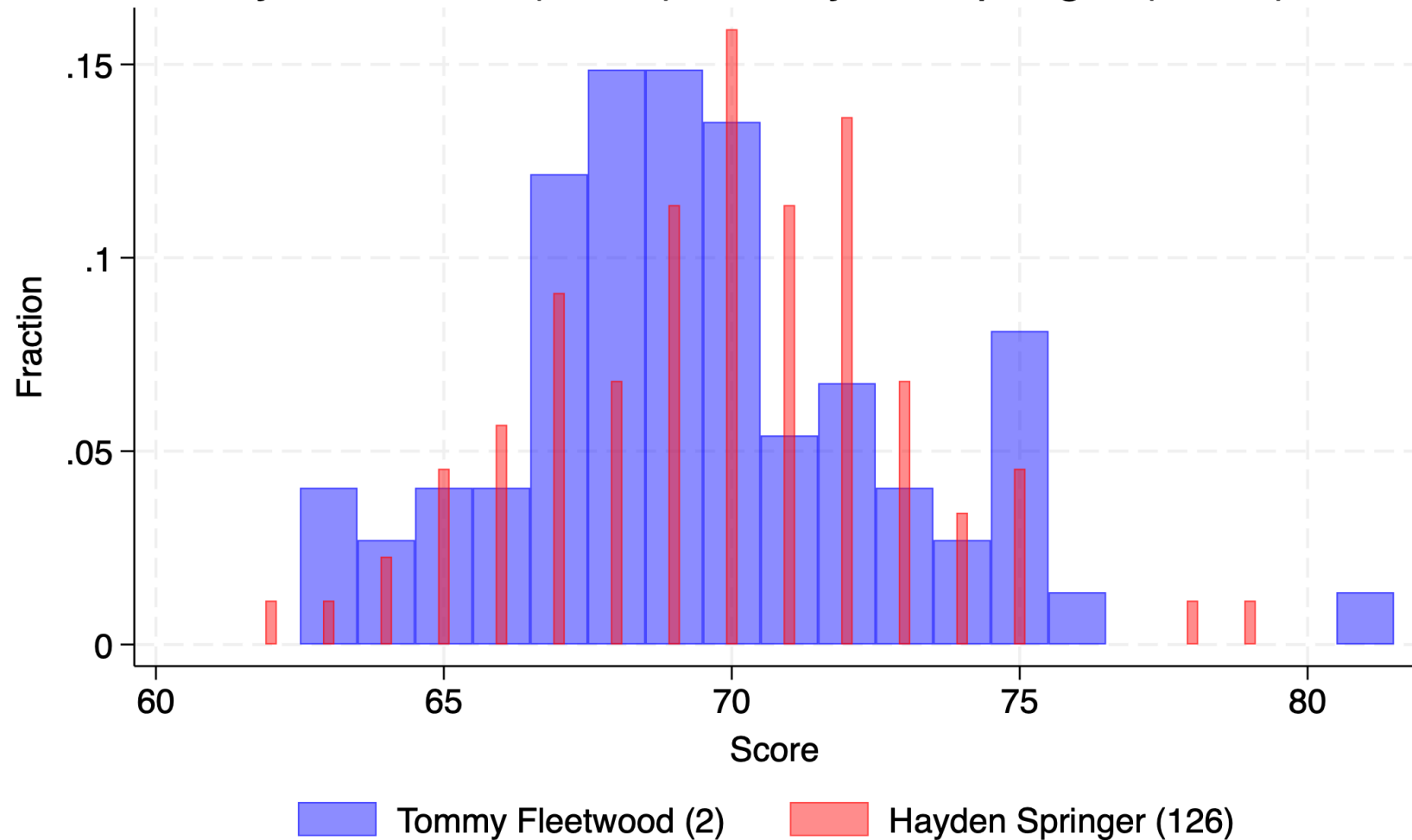
Signal vs. Noise in Golf

Scottie Scheffler (67.88) vs. Hayden Springer (69.86), 2025



Signal vs. Noise in Golf

Tommy Fleetwood (69.43) vs. Hayden Springer (69.86), 2025



Hot Hand (Tversky-Gilovich, 1989)

- What is the “hot hand”?
 - “If I’m on, I find that confidence just builds in you. You feel nobody can stop you. It’s important to hit that first one especially if it’s a wish. Then you hit another, and your mind goes buggy. You feel like you can do anything.”

— World B. Free



“Success breeds success; failure breeds failure”

Hot Hand: Study 1

2017-2018 Golden State Warriors, Field Goals

Definition of Hot Hand 1: “A player has a better chance of making a shot after having just made his last two or three shots than he does after having just missed his last two or three shots.” (91% agree)

Player	$P(H HHH)$	$P(H HH)$	$P(H H)$	$P(H)$	n	$P(H M)$	$P(H MM)$	$P(H MMM)$	Serial Corr
Andre Iguodala	0.471	0.460	0.500	0.463	320	0.433	0.438	0.415	0.067
David West	0.536	0.548	0.583	0.571	378	0.553	0.549	0.531	0.031
Draymond Green	0.473	0.437	0.448	0.454	619	0.460	0.497	0.473	-0.012
JaVale McGee	0.697	0.733	0.662	0.621	219	0.549	0.568	0.438	0.113
Jordan Bell	0.592	0.645	0.655	0.627	185	0.574	0.571	0.500	0.081
Kevin Durant	0.513	0.472	0.505	0.516	1,221	0.527	0.527	0.508	-0.022
Kevon Looney	0.469	0.516	0.559	0.583	192	0.613	0.613	0.583	-0.054
Klay Thompson	0.504	0.498	0.471	0.488	1,179	0.503	0.517	0.531	-0.031
Nick Young	0.324	0.435	0.423	0.412	488	0.406	0.385	0.404	0.017
Omri Casspi	0.576	0.523	0.555	0.580	207	0.621	0.576	0.571	-0.066
Patrick McCaw	0.143	0.212	0.359	0.407	226	0.444	0.411	0.452	-0.085
Quinn Cook	0.583	0.537	0.554	0.484	252	0.415	0.461	0.537	0.138
Shaun Livingston	0.400	0.427	0.477	0.502	343	0.524	0.563	0.618	-0.047
Stephen Curry	0.409	0.441	0.493	0.495	864	0.499	0.477	0.491	-0.006
Zaza Pachulia	0.551	0.590	0.553	0.566	265	0.588	0.630	0.688	-0.034
Golden State Warriors	0.483	0.498	0.520	0.518	6,958	0.514	0.519	0.516	0.006

Hot Hand: Study 1

2017-2018 Cleveland Cavaliers, Field Goals

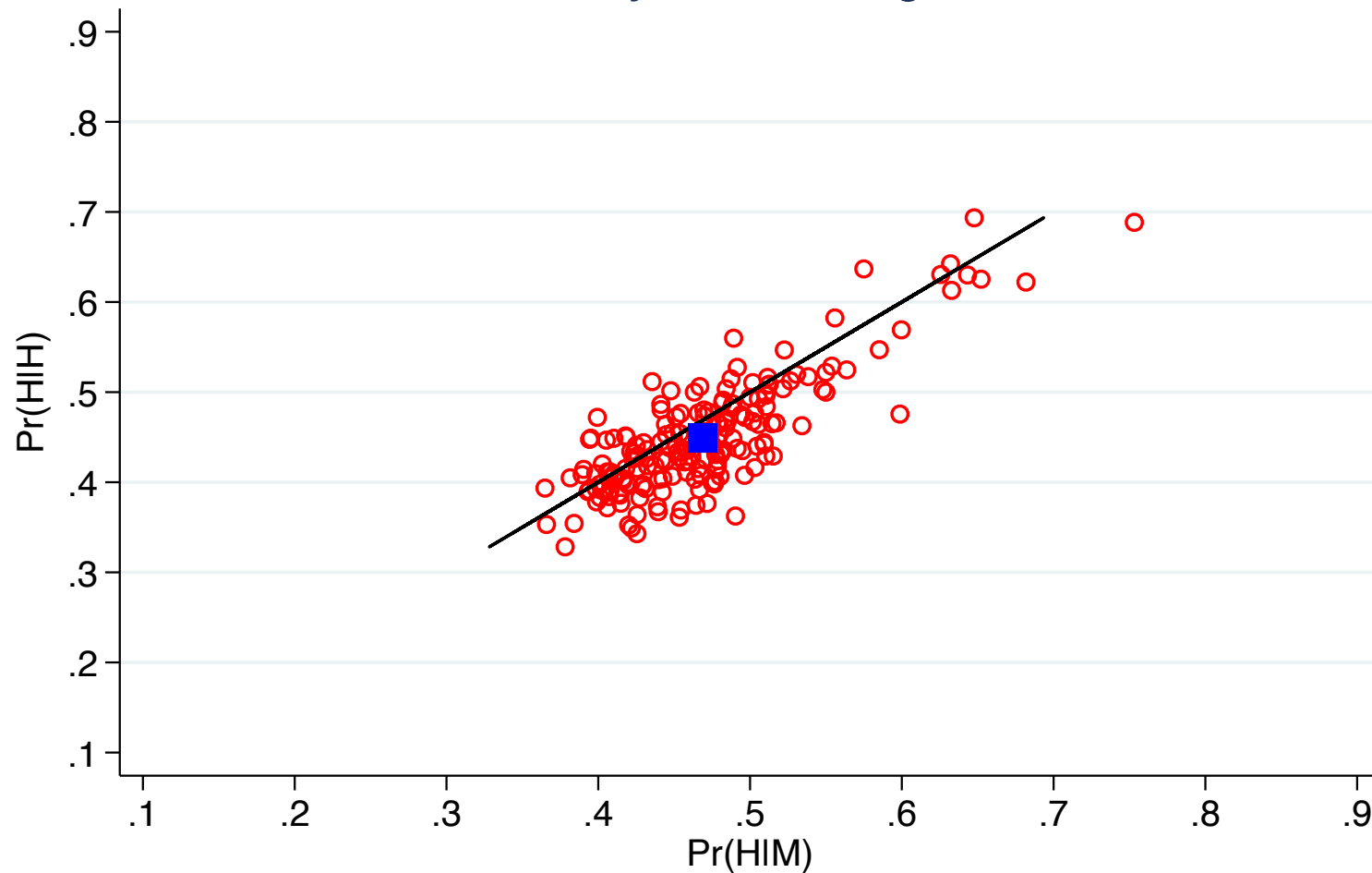
Player	P(H HHH)	P(H HH)	P(H H)	P(H)	n	P(H M)	P(H MM)	P(H MMM)	Serial Corr
Cedi Osman	0.444	0.400	0.500	0.484	186	0.474	0.480	0.462	0.026
Channing Frye	0.353	0.447	0.452	0.497	169	0.536	0.564	0.471	-0.083
Derrick Rose	0.417	0.480	0.439	0.439	132	0.446	0.475	0.476	-0.007
Dwyane Wade	0.525	0.455	0.444	0.453	437	0.458	0.449	0.449	-0.014
George Hill	0.467	0.395	0.458	0.444	187	0.433	0.475	0.484	0.025
Isaiah Thomas	0.600	0.440	0.377	0.361	194	0.355	0.338	0.302	0.022
JR Smith	0.378	0.446	0.423	0.403	595	0.392	0.398	0.415	0.031
Jae Crowder	0.368	0.352	0.355	0.418	376	0.459	0.492	0.459	-0.104
Jeff Green	0.508	0.452	0.458	0.477	619	0.495	0.509	0.413	-0.038
Jordan Clarkson	0.381	0.379	0.454	0.456	287	0.459	0.482	0.523	-0.005
Jose Calderon	0.577	0.553	0.505	0.503	185	0.506	0.386	0.444	0.000
Kevin Love	0.397	0.420	0.449	0.458	730	0.466	0.493	0.467	-0.017
Kyle Korver	0.342	0.406	0.455	0.459	484	0.464	0.468	0.446	-0.009
Larry Nance Jr.	0.478	0.500	0.552	0.550	160	0.548	0.546	0.667	0.004
LeBron James	0.515	0.518	0.527	0.542	1,580	0.562	0.580	0.594	-0.035
Rodney Hood	0.385	0.293	0.455	0.442	199	0.432	0.524	0.567	0.022
Tristan Thompson	0.535	0.589	0.553	0.559	236	0.563	0.467	0.417	-0.010
Cleveland Cavaliers	0.451	0.443	0.462	0.467	6,756	0.473	0.478	0.474	-0.011

2017-2018 All NBA Players, 500+ Field Goal Attempts

All Players (500+ Field Goal Attempts)	0.442	0.449	0.454	0.461	148,163	0.468	0.471	0.475	-0.014
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Hot Hand: Study 1

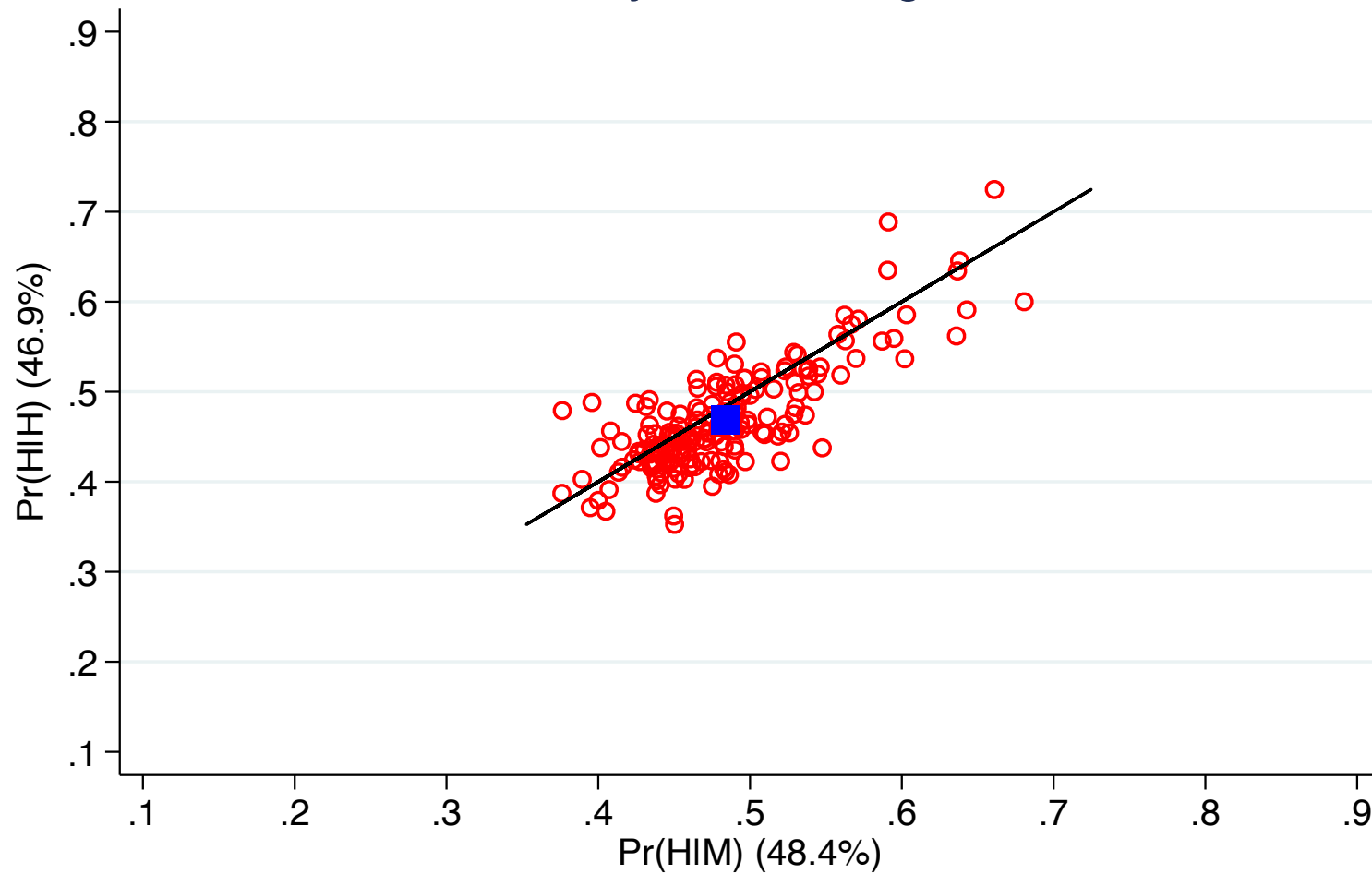
Hot Hand analysis: Looking back 1 shot



NBA 2021-2022, players with 500+ field goal attempts

Hot Hand: Study 1

Hot Hand analysis: Looking back 1 shot



NBA 2023-2024, players with 500+ field goal attempts

Hot Hand: Study 2

2017-2018 Golden State Warriors, Free Throws

Definition of Hot Hand 2: “When shooting free throws, a player has a better chance of making his second shot after making his first shot.” (68% agree)

	$P(H2 H1)$	$P(H)$	n	$P(H2 M1)$	serial correlation
Andre Iguodala	0.652	0.645	76	0.733	-0.085
David West	0.913	0.779	68	0.818	0.138
Draymond Green	0.780	0.782	156	0.895	-0.125
Kevin Durant	0.899	0.875	321	0.870	0.033
Kevon Looney	0.500	0.571	70	0.539	-0.037
Klay Thompson	0.879	0.842	82	0.875	0.005
Omri Casspi	0.810	0.746	59	0.667	0.155
Stephen Curry	0.921	0.929	211	0.833	0.073
Zaza Pachulia	0.735	0.798	79	0.833	-0.081
Golden State	0.788	0.774	1122	0.785	0.008

	$P(H2 H1)$	$P(H)$	n	$P(H2 M1)$	serial correlation
All Players (100+ Free throws)	0.799	0.771	30776	0.777	0.023

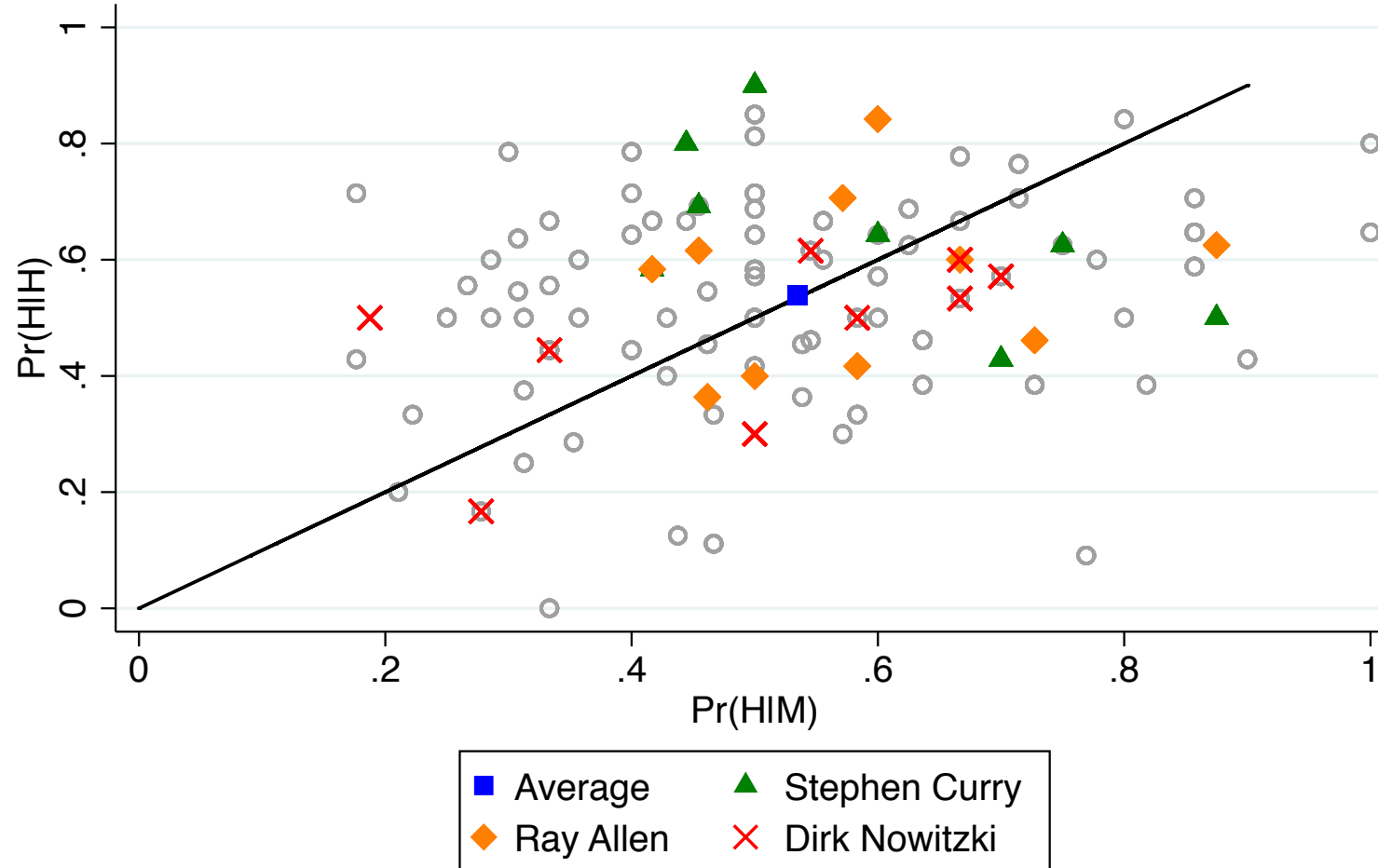
Hot Hand: Study 3

NBA Long Distance Shooting Contest, 25
shots/round, 2000-2016

	$P(H2 H1)$	$P(H)$	n	$P(H2 M1)$	serial correlation
Booker (Round 1)	0.500	0.520	25	0.583	-0.084
Booker (Round 2)	0.385	0.520	25	0.727	-0.343
Curry (Round 1)	0.800	0.640	25	0.444	0.365
Curry (Round 2)	0.625	0.680	25	0.750	-0.125
Harden (Round 1)	0.615	0.560	25	0.546	0.071
K. Thompson (Round 1)	0.706	0.720	25	0.714	-0.008
K. Thompson (Round 2)	0.778	0.760	25	0.667	0.111
Lowry (Round 1)	0.500	0.480	25	0.500	0.000
McCollum (Round 1)	0.500	0.400	25	0.286	0.218
Middleton (Round 1)	0.500	0.400	25	0.357	0.143
Redick (Round 1)	0.667	0.600	25	0.444	0.218
All Players (2016)	0.598	0.571	275	0.547	0.052
All Players (2000-2016)	0.539	0.528	3,375	0.535	0.004

Hot Hand: Study 3

Hot Hand analysis: NBA 3-point shooting contest



Note: 2000-2016 contests (n=159 player-year-rounds)

Hot Hand and Streaks

Simulation of 25 shots

- 5 Possible Cases:

$$P(H_{i+1}|H_i) = P(M_{i+1}|M_i) =$$

10%, 30%, 50%, 70%, 90%

- Which is which?

- All combinations are possible

Observation	Series				
	1	2	3	4	5
1	M	M	H	M	H
2	M	H	H	H	M
3	H	H	M	M	H
4	M	H	H	H	M
5	M	M	M	H	H
6	H	H	H	H	M
7	M	M	M	H	M
8	M	H	H	H	H
9	M	H	M	M	M
10	H	H	H	M	M
11	M	M	M	M	M
12	H	H	H	H	H
13	M	M	M	H	M
14	H	M	H	M	H
15	H	H	M	M	H
16	M	M	H	M	M
17	M	H	M	H	H
18	M	H	H	H	M
19	M	H	M	H	H
20	M	M	H	M	M
21	H	H	H	H	H
22	H	H	M	H	M
23	H	M	H	M	H
24	M	M	M	M	M
25	H	H	H	H	M
Guess?					

		Series				
	Observation	1	2	3	4	5
	1	0	0	1	0	1
	25	1	1	1	1	0
	True Process	0.5	0.3	0.1	0.5	0.3
BAYES						
DATA						

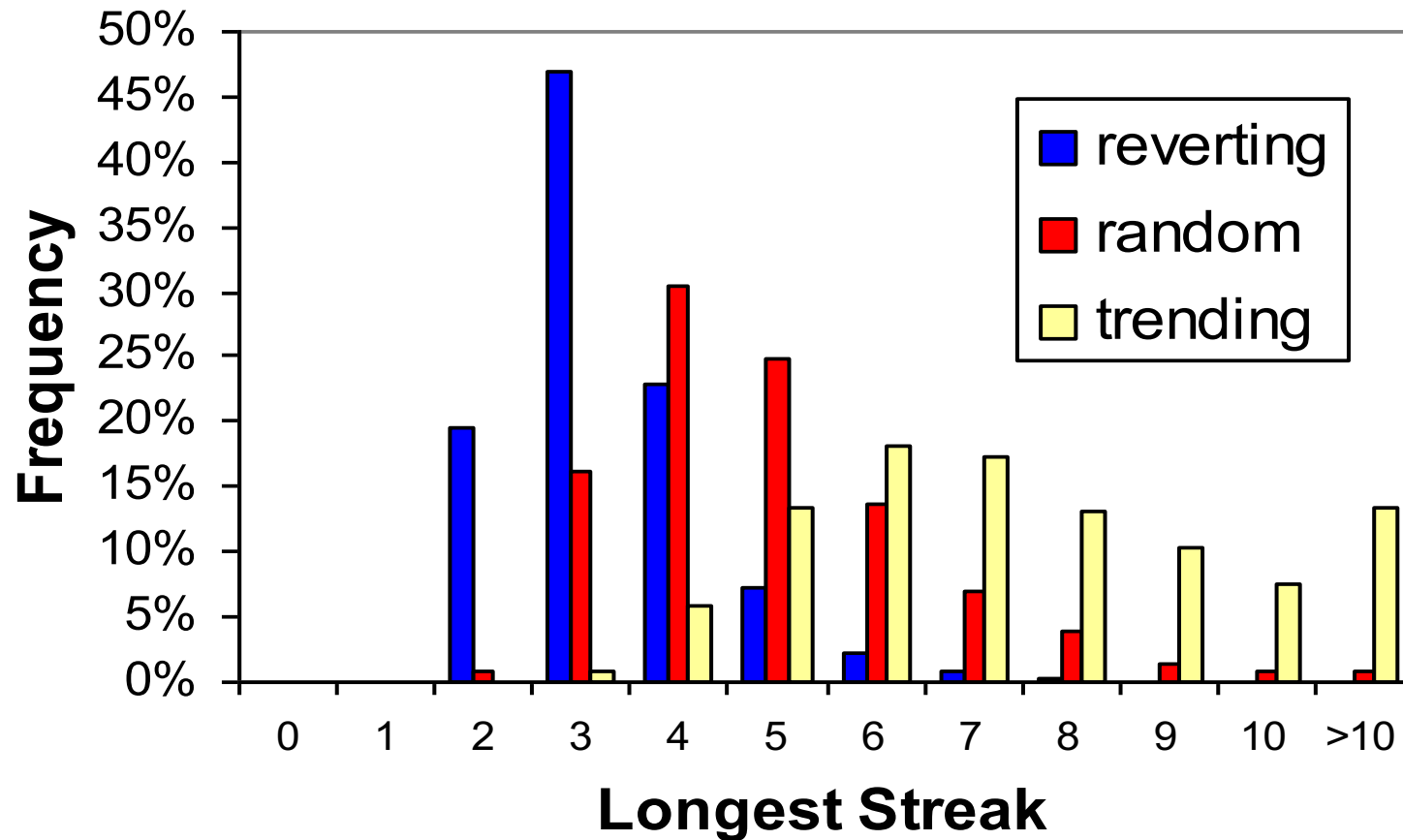
Hot Hand and Streaks I

$$P(H_{i+1}|H_i)=P(M_{i+1}|M_i)$$

Shot	Made=1, Missed=0								
1	1	0	0	1	1	1	1	1	0
2	0	1	0	0	0	1	1	1	1
3	0	0	1	1	0	0	1	1	1
4	1	1	0	1	0	0	1	0	1
5	0	0	1	0	1	1	1	0	1
6	1	1	0	1	1	0	1	0	1
7	0	0	1	1	1	0	1	0	1
8	1	1	0	1	1	0	1	0	1
9	0	0	0	0	1	0	1	1	1
10	1	1	1	1	0	0	1	1	1
11	0	0	0	0	1	1	1	1	1
12	1	1	1	1	0	1	1	1	1
13	0	1	1	0	1	1	1	1	1
14	0	1	0	1	0	0	0	1	1
15	1	0	1	0	1	0	0	1	1
16	0	1	0	1	1	0	0	1	1
17	1	0	1	1	0	1	0	1	1
18	0	1	0	0	1	0	0	1	1
19	1	0	1	1	0	0	0	1	0
20	0	1	0	0	1	1	0	1	0
21	1	1	1	1	1	1	1	1	0
22	0	0	0	0	0	1	0	1	0
23	1	1	1	0	1	1	0	1	0
24	0	1	0	0	1	1	0	1	0
25	1	0	0	0	0	1	1	1	0

Hot Hand and Streaks II

Simulation of 25 shots: $P(H_{i+1}|H_i) = .3, .5, .7$ ($n=2500$)



Why do we retain incorrect beliefs?

- We have mistaken models of chance processes
 - Events that seem surprising are not statistically surprising (e.g., regression to the mean; perceptions of chance processes)
- We prefer causal stories to chance explanations
 - We prefer to see the world as orderly and systematic
 - We have great facility at explaining any “puzzle”
- We like to believe what we want to believe
 - Self-serving beliefs maintain self-esteem
 - Positive Illusion (Class 1)
 - People tend to attribute success to ability and failure to bad luck (Illusion of Control)
 - ... but individuals believe they are being objective
- Feedback/evidence are ambiguous and incomplete
 - Example 1: Football, shot off crossbar
 - Example 2: An employee volunteers to stay late (What attribution?)

Exercise

- ❑ Rule discovery: “2, 4, 6” is a triple that satisfies an unarticulated rule. What is the rule?
- ❑ Task: Give me other triples, and I will tell you whether these triples satisfy the rule or not.

Wason Test in the New York Times

<http://www.nytimes.com/interactive/2015/07/03/upshot/a-quick-puzzle-to-test-your-problem-solving.html>

TheUpshot

EDITED BY DAVID LEONHARDT
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A Quick Puzzle to Test Your Problem Solving

By DAVID LEONHARDT and YOU JULY 2, 2015

A short game sheds light on government policy, corporate America and why no one likes to be wrong.

Here's how it works:

We've chosen a rule that some sequences of three numbers obey — and some do not. Your job is to guess what the rule is.

We'll start by telling you that the sequence 2, 4, 8 obeys the rule:

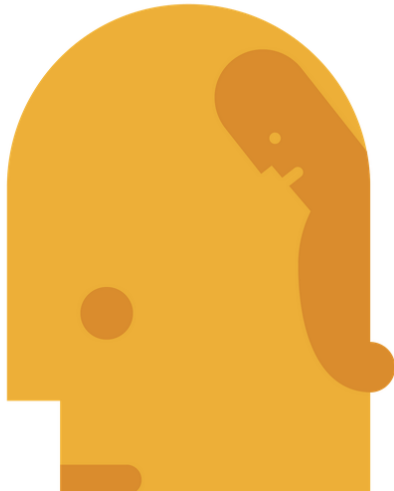
2

4

8

Obeys the rule

Now it's your turn. Enter a number sequence in the boxes below, and we'll tell you whether it satisfies the rule or not. You can test as many sequences as you want.



- “Remarkably, 78 percent of people who have played this game so far have guessed the answer without first hearing a single no. A mere 9 percent heard at least three nos — even though there is no penalty or cost for being told no...” David Leonhardt, *New York Times*

Confirmation Bias

The tendency to ask questions or seek information that is consistent with a prior hypothesis (and to avoid asking questions or seeking information that might disconfirm a favored hypothesis).

“They still cling stubbornly to the idea that the only good answer is a yes answer. If they say, ‘Is the number between 5,000 and 10,000?’ and I say yes, they cheer; if I say no, they groan, even though they get exactly the same amount of information in either case.”

John Holt, *Why Children Fail*

“The human understanding, when any proposition has been once laid down...forces everything else to add fresh support and confirmation...it is the peculiar and perpetual error of the human understanding to be more moved and excited by affirmatives than negatives.”

Francis Bacon (1620)

Confirmation Biases

■ Types

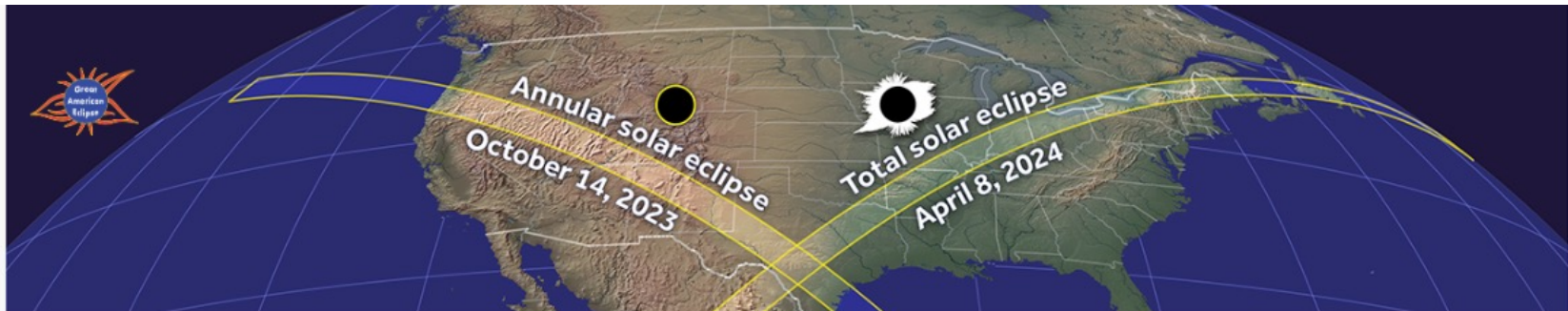
- ❑ Seek information that confirms (both purposely and non-purposely)...
- ❑ Interpret ambiguous information as consistent with...
- ❑ Operate in groups/networks that support...
...a prior hypothesis.

What role did Confirmation Bias play at AOL?



Reflect on a recent decision. Did confirmation bias play a role in in what information was collected? How information was weighted?

April 8, 2024



[TOTALITY](#) / [ECLIPSE DAILY](#) / [2024 TOTAL](#) / [2023 ANNULAR](#) / [21ST CENTURY](#) / [2017 TOTAL](#) / [HISTORY](#) / [STORE](#) / [MORE](#) / [SIGN IN](#)

Fly over the 2024 total solar eclipse

This animation of the Moon's shadow across North America simulates the view from a spacecraft 125 miles high chasing the shadow! On April 8, 2024, everyone inside the shadow will experience the breathtaking sight of the Sun's corona during the total solar eclipse. Totality races from Mazatlan to Newfoundland.

Email Address (required)

Newsletter updates with timely information on solar eclipses around the world. We never sell or misuse email addresses.

Submit

Total Solar Eclipse April 8, 2024

Time of totality

11:06:50 am MST

Duration of totality

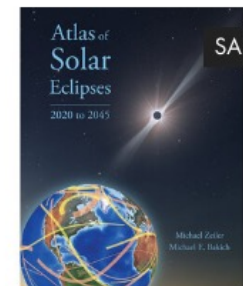
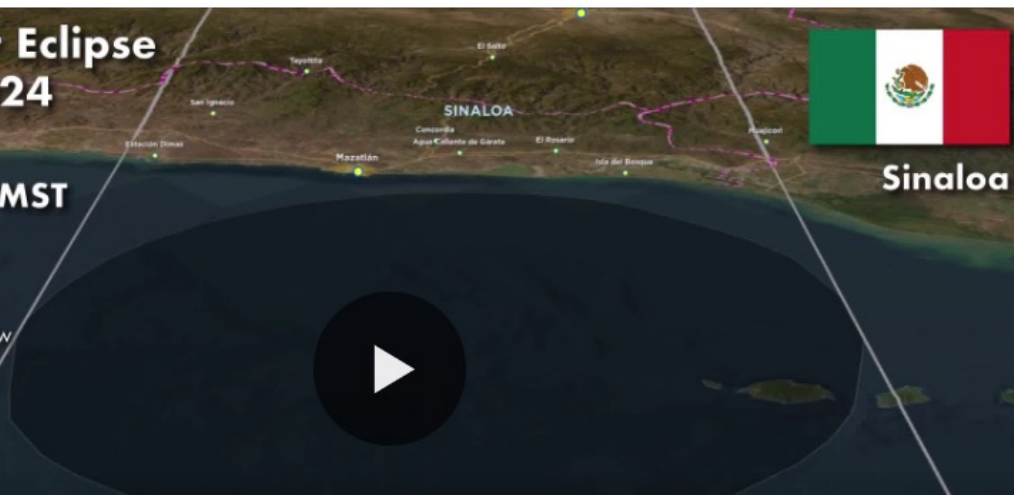
4 min 26 sec

Speed of Moon's shadow

1,564 mph

Width of path of totality

123.4 miles



**Atlas of Solar Eclipses
- 2020 to 2045**

\$50.00 \$75.00

August 8, 1999 (Karlsruhe)



Current Explanation

“This celestial event is a solar eclipse in which the moon passes between the sun and Earth and blocks all or part of the sun for up to about three hours, from beginning to end, as viewed from a given location. For this eclipse, the longest period when the moon completely blocks the sun from any given location along the path will be about two minutes and 40 seconds. The last time the contiguous U.S. saw a total eclipse was in 1979.”

<https://eclipse2017.nasa.gov/eclipse-who-what-where-when-and-how>

Pre-scientific explanations

- Sun is attacked by evil mythological creatures
 - Dragon (China)
 - Werewolf (Romania)
 - Giant turtle (Vietnam)
 - Jaguar (Latin America)
 - *You need to save the sun!*
 - China: scare away the dragon (shoot fireworks, beats gongs, etc.)
- Solar eclipses cause birth problems
 - Mexico: Pregnant women went outside with belt around their waste fastened with red arrowhead
- Solar eclipses are a poison
 - India: 12 hours of fasting
- How do you disconfirm?

Example 1: CEOs and their Pets

- From *Management Focus* Magazine:

“Results of a recent survey of 74 chief executive officers indicate that there may be a link between childhood pet ownership and future career success.

Fully 94% of the CEOs, all of them employed within Fortune 500 companies, had possessed a dog, a cat, or both as youngsters....

The respondents asserted that pet ownership had helped them to develop many of the positive character traits that make good managers today, including responsibility, empathy, respect for other living beings, generosity, and good communication skills.”

Example 2

- The decision aid to rule them all!



Example 2: Google Searches

<http://rationallyspeaking.blogspot.co.uk/2010/01/is-google-making-us-less-rational.html?m=1>



The term "confirmation bias" was first used in a classic 1960 paper by P.C. Wason called [On the failure to eliminate hypotheses in a conceptual task](#). Subjects were given a sequence of numbers: "2,4,6" and told that the numbers were picked according to a rule, and the goal was for the subjects to guess that rule. They could test their guesses by suggesting other sequences and the experimenter would answer "Yes, that fits the rule" or "No, that doesn't fit the rule."

Example 2: Google Searches

Hypothesis: “Obama born in Kenya.”

“Obama born in Kenya”

“Obama born in US”

Barack Obama Admits he was born in Kenya Speech - YouTube



<https://www.youtube.com/watch?v=cQcd41RO25k> ▾
Feb 28, 2014 - Uploaded by skin88p
Barack Obama Admits he was born in Kenya Speech.

Barack Obama citizenship conspiracy theories - Wikipedia

https://en.wikipedia.org/wiki/Barack_Obama_citizenship_conspiracy_theories ▾
... House website, reaffirming that he was born on August 4, 1961, in Honolulu, Hawaii. During Barack Obama's campaign for president in 2008 and in the years following his election, From the start of March 2008, rumors that Obama was born in Kenya before being flown to Hawaii were spread on conservative websites, ...

Citizenship Scandal: 7 Facts About Barack Obama's Birth Certificate ...

www.newsmax.com/FastFeatures/Barack-Obama-Citizenship...Birth.../01/.../621307/ ▾
Jan 28, 2015 - According to The Huffington Post, even if Obama had been born in Kenya, "the president still would have been a U.S. citizen at birth, because ...

PolitiFact statements about Obama Birth Certificate

www.politifact.com/subjects/obama-birth-certificate/ ▾
Recent statements involving Obama Birth Certificate. False. Despite ... "I finished" the controversy about where President Barack Obama was born. — PolitiFact ...

Barack Obama / Born

August 4, 1961 (age 55 years), Kapiolani Medical Center for Women and Children, Honolulu, HI



Donald Trump
June 14,
1946



Hillary Clinton
October 26,
1947



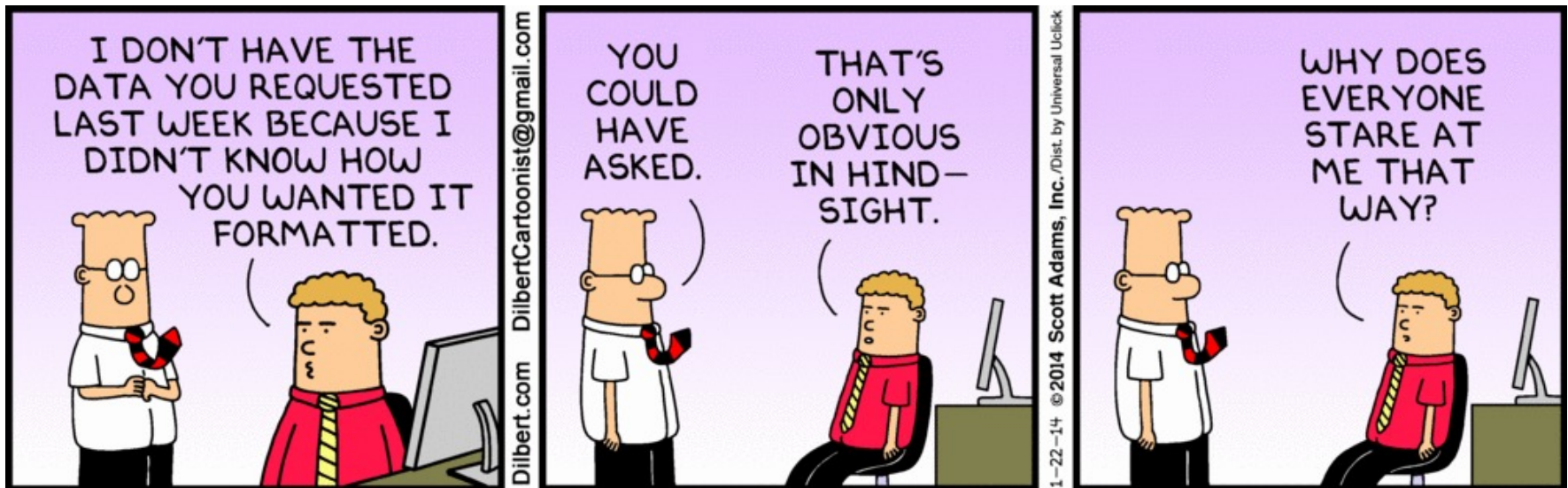
Michelle Obama
Spouse
January 17,
1964

Feedback

Example 2: Google Searches

Topic	Term 1	Confirming 1	Term 2	Confirming 2
Birther	"Obama birth certificate real"	70%	"Obama birth certificate fake"	80%
Vaccination and autism	"vaccination causes autism"	60%	"vaccination doesn't cause autism"	90%
moon landing	"moon landing real"	80%	"moon landing fake"	50%
God	"Gods exists"	90%	"God doesn't exist"	100%
marriage	"Should I get married"	75%	"Should I not get married"	79%
Coke vs. Pepsi	"Pepsi is better than Coke"	83%	"Coke is better than Pepsi"	86%
Apple stock	"Should I buy Apple stock"	69%	"Should I sell Apple stock"	86%
Holocaust	"Holocaust was fake"	90%	"Holocaust was fake debunked"	100%

Hindsight bias

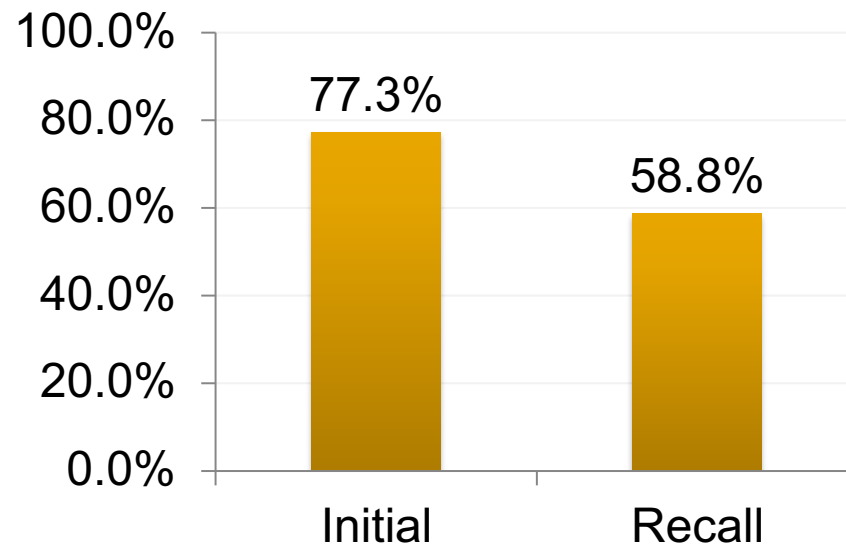


Hindsight Bias I

- Hindsight bias (a.k.a. “Monday morning Quarterback”)
 - Tendency to overestimate the chance that we would have predicted the result (or known the answer)

Hindsight Bias II

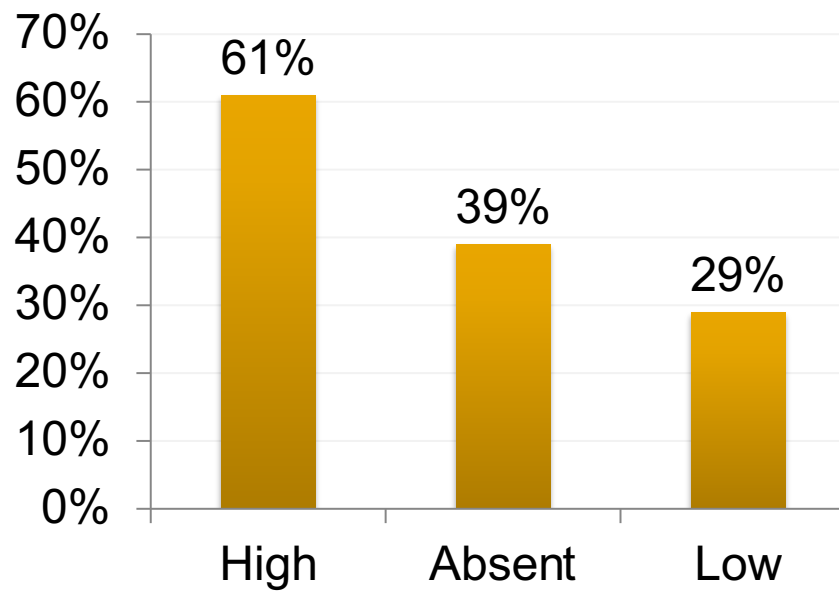
- Example 2 (Cassair and Craig, 2009)
 - Study of 198 failed entrepreneurs
 - Likelihood that startup would become an operating business
 - Initial
 - Likelihood that startup would become an operating business
 - After failure
 - Recall initial likelihood



Hindsight bias is not influenced by startup experience of entrepreneur!

Hindsight Bias III

- Example 3 (Bukzar and Connolly, 1988)
 - MBA students prepared a case about pharmaceutical firm expansion
 - Success Criteria: 20% ROI after 2 years
 - First year feedback: High (+36%); Absent (--); Low (+4%)
 - Recall prediction of success



Hindsight Bias IV

- Example 4 (Thaler, Managerial Decision Making-Cornell, 1986)
 - “On the first day of class you were asked to estimate the probability that certain events would happen between February 1 and April 1, 1986. For each of these events, listed below, please indicate first whether you believe the event did or did not happen (if are not sure, guess) and then the probability you remember assigning the first day of class. The person who comes closest to correctly remembering his or her own original probabilities will be given a prize of \$10.”

Hindsight Bias and Organizations

- What role does hindsight bias play in organizational decisions?

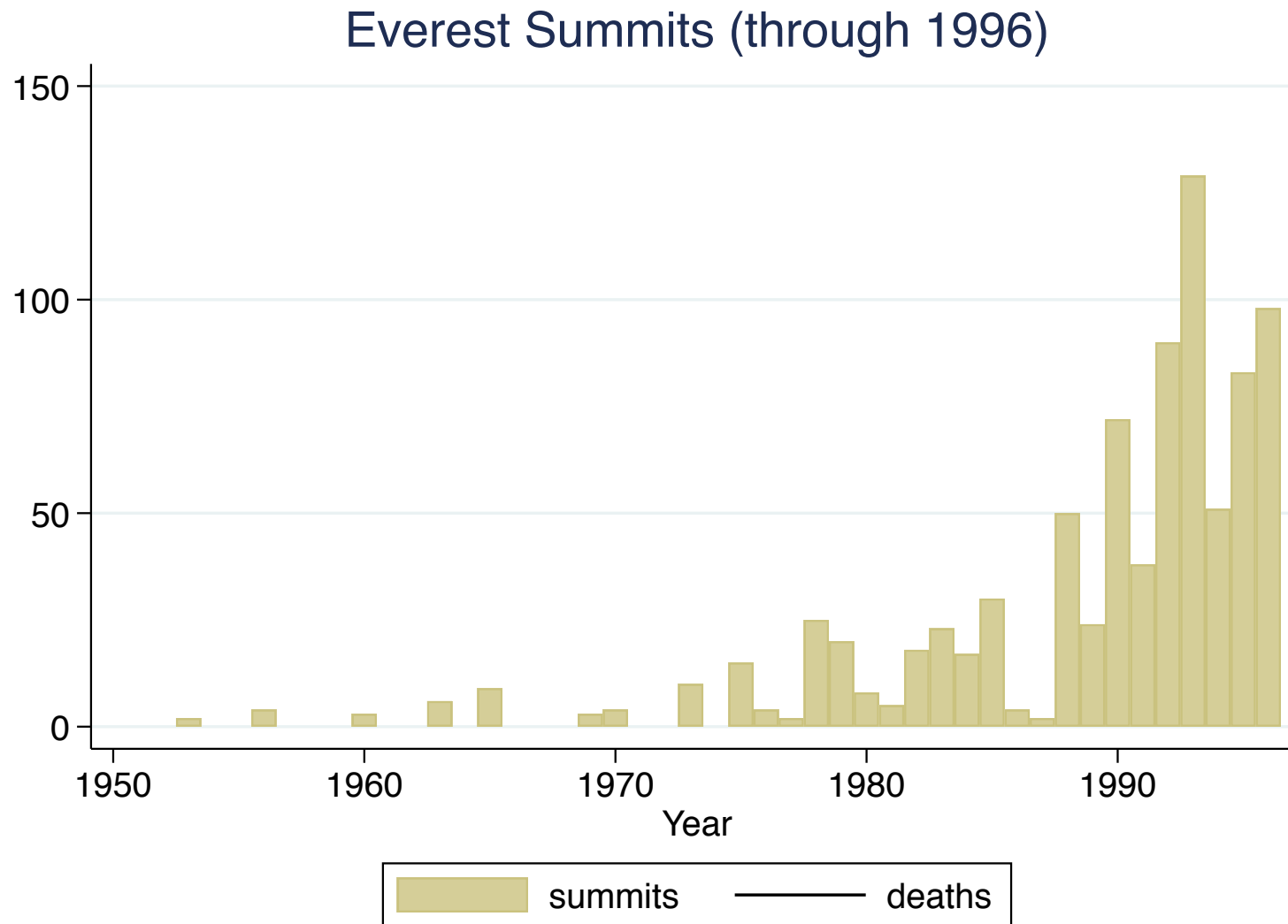
The Everest Challenge

But there are men for whom the unattainable has a special attraction. Usually they are not experts: their ambitions and fantasies are strong enough to brush aside the doubts which more cautious men might have. Determination and faith are their strongest weapons. At best such men are regarded as eccentric; at worst, mad . . .

Everest has attracted its share of men like these. Their mountaineering experience varied from none at all to very slight—certainly none of them had the kind of experience which would make an ascent of Everest a reasonable goal. Three things they all had in common: faith in themselves, great determination, and endurance.

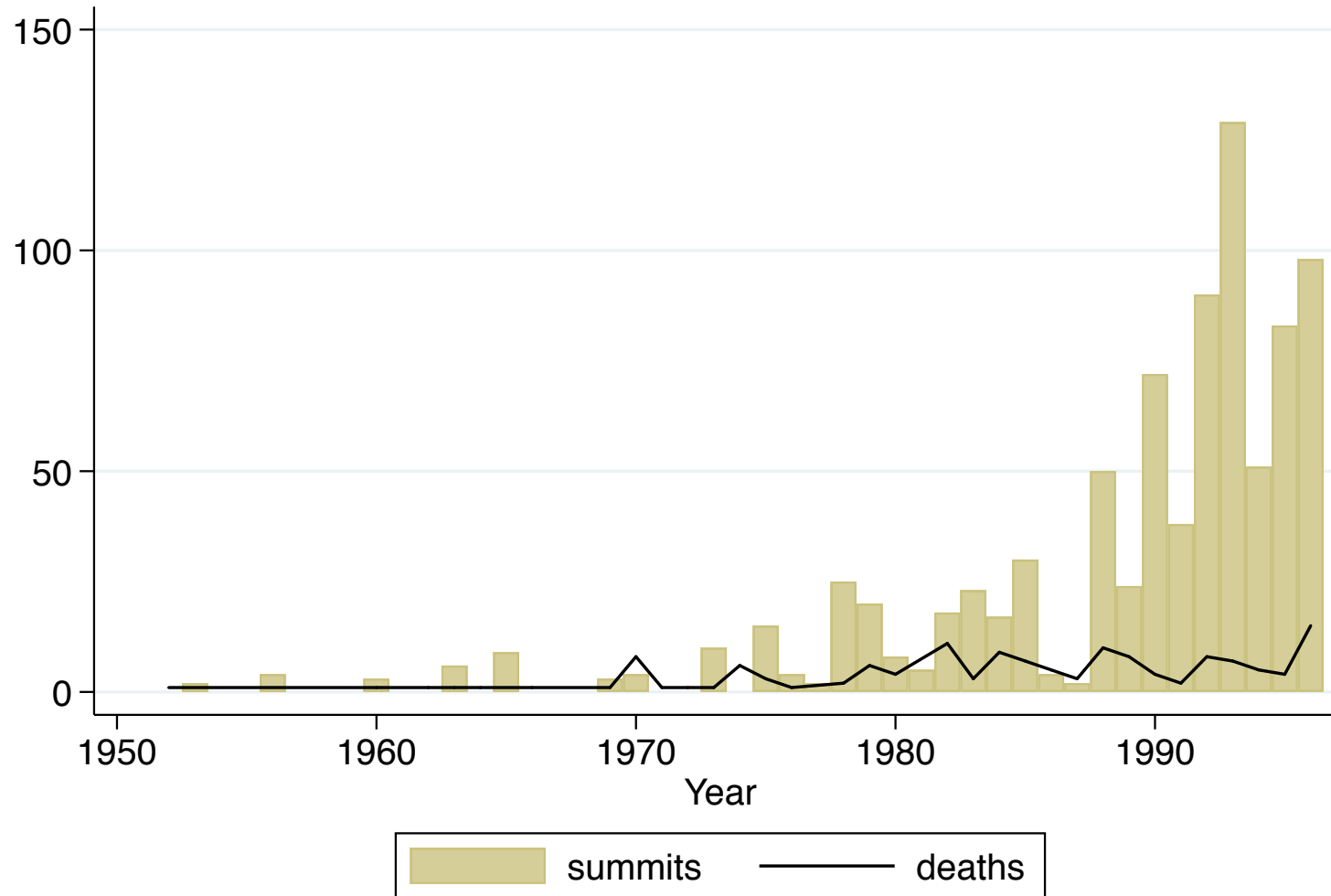
Walt Unsworth
Everest

Everest Summits



Everest Summits (and Deaths)

Everest Summits (through 1996)



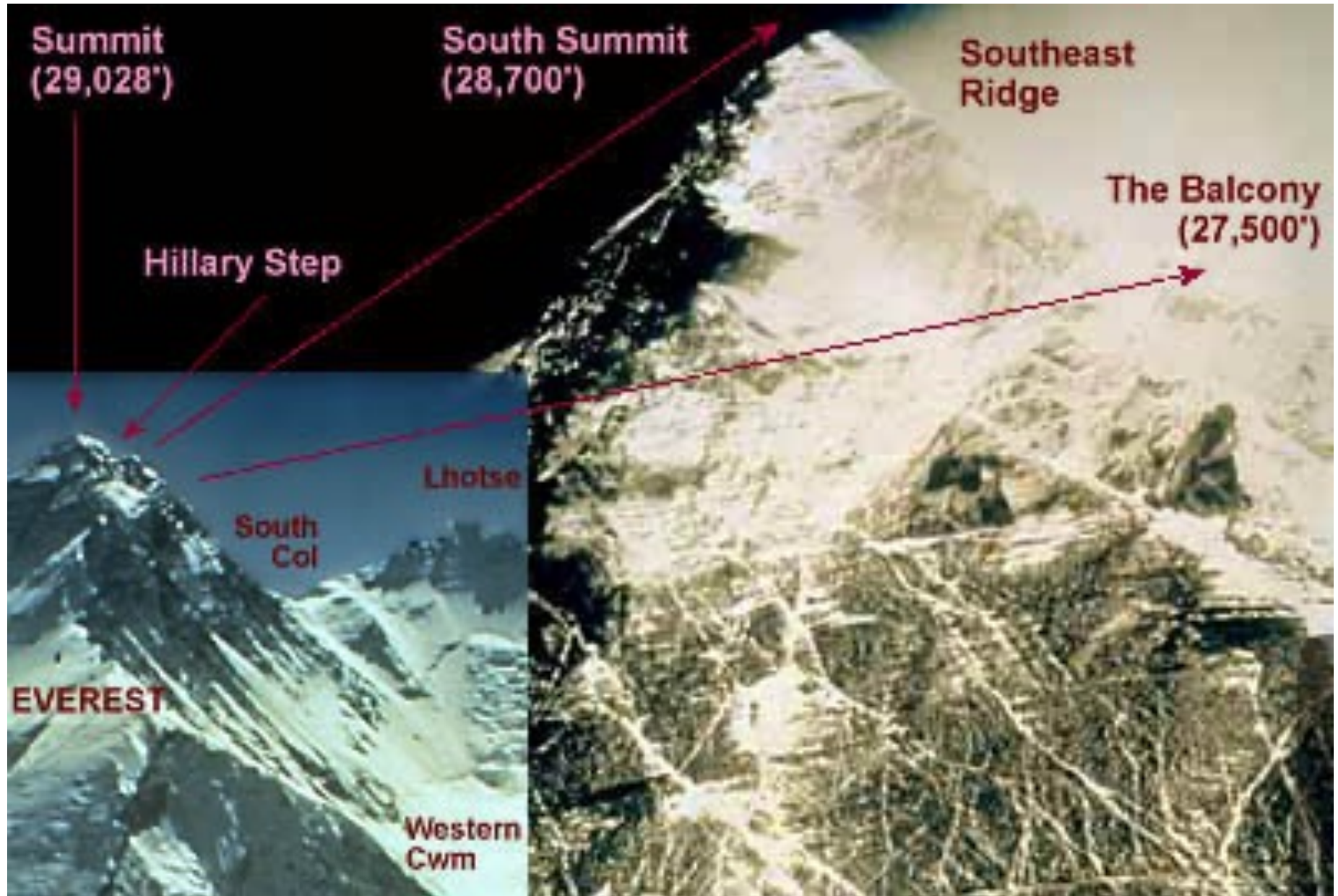
Everest Summit



Khumbu Icefall



Everest Summit II



Hillary Step

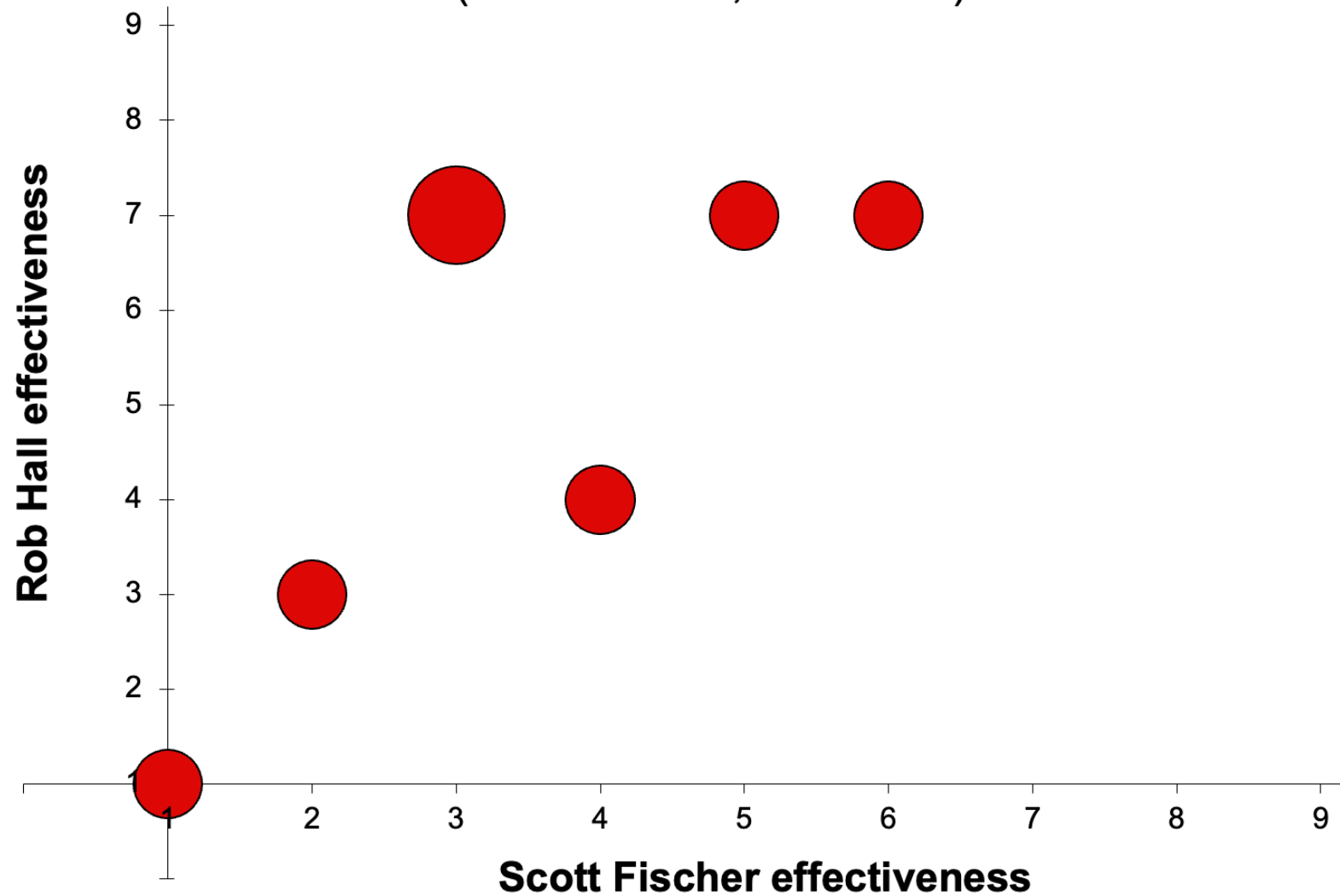


Hillary Step (May 19, 2012)

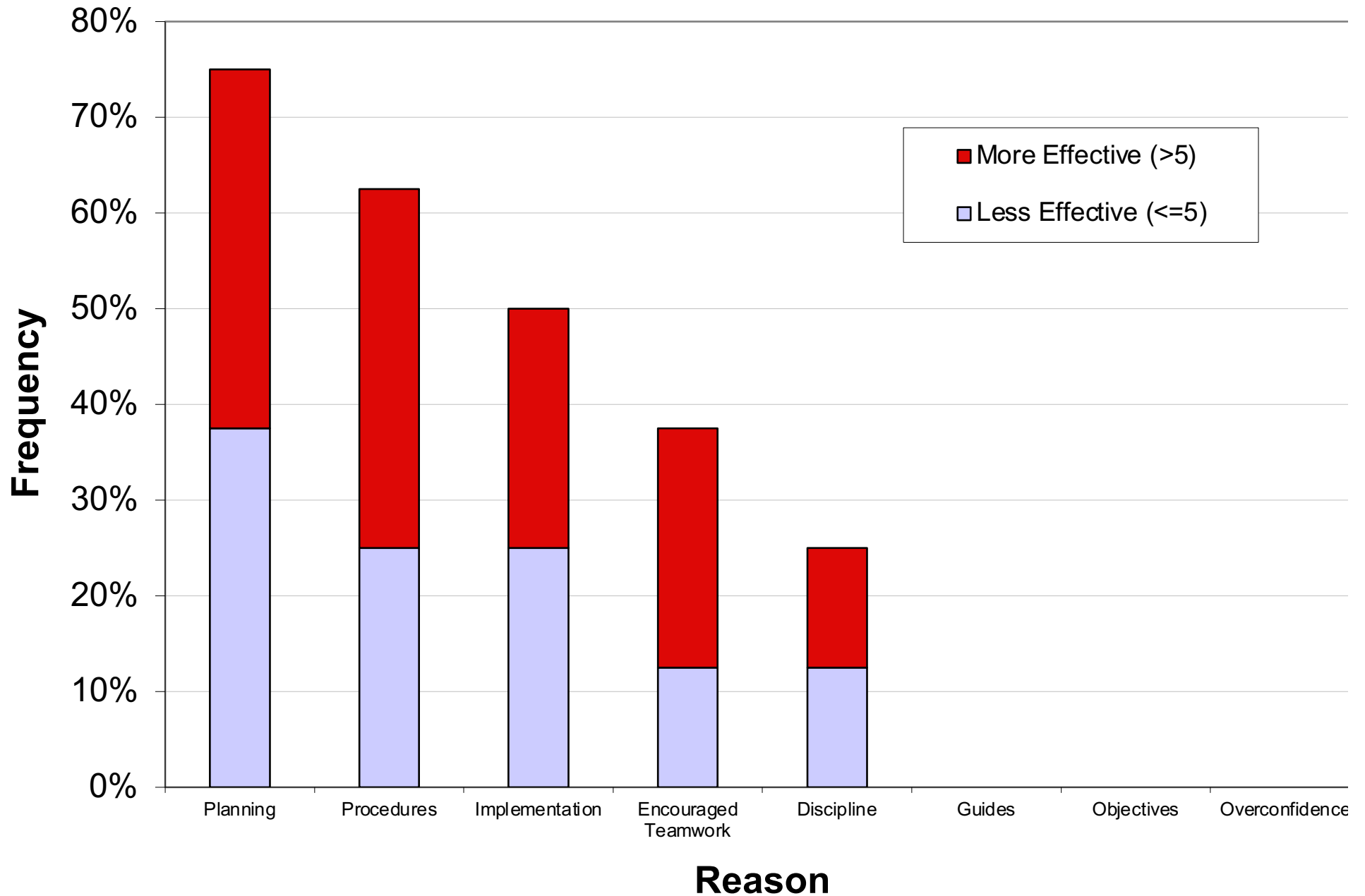


Management Philosophy Effectiveness

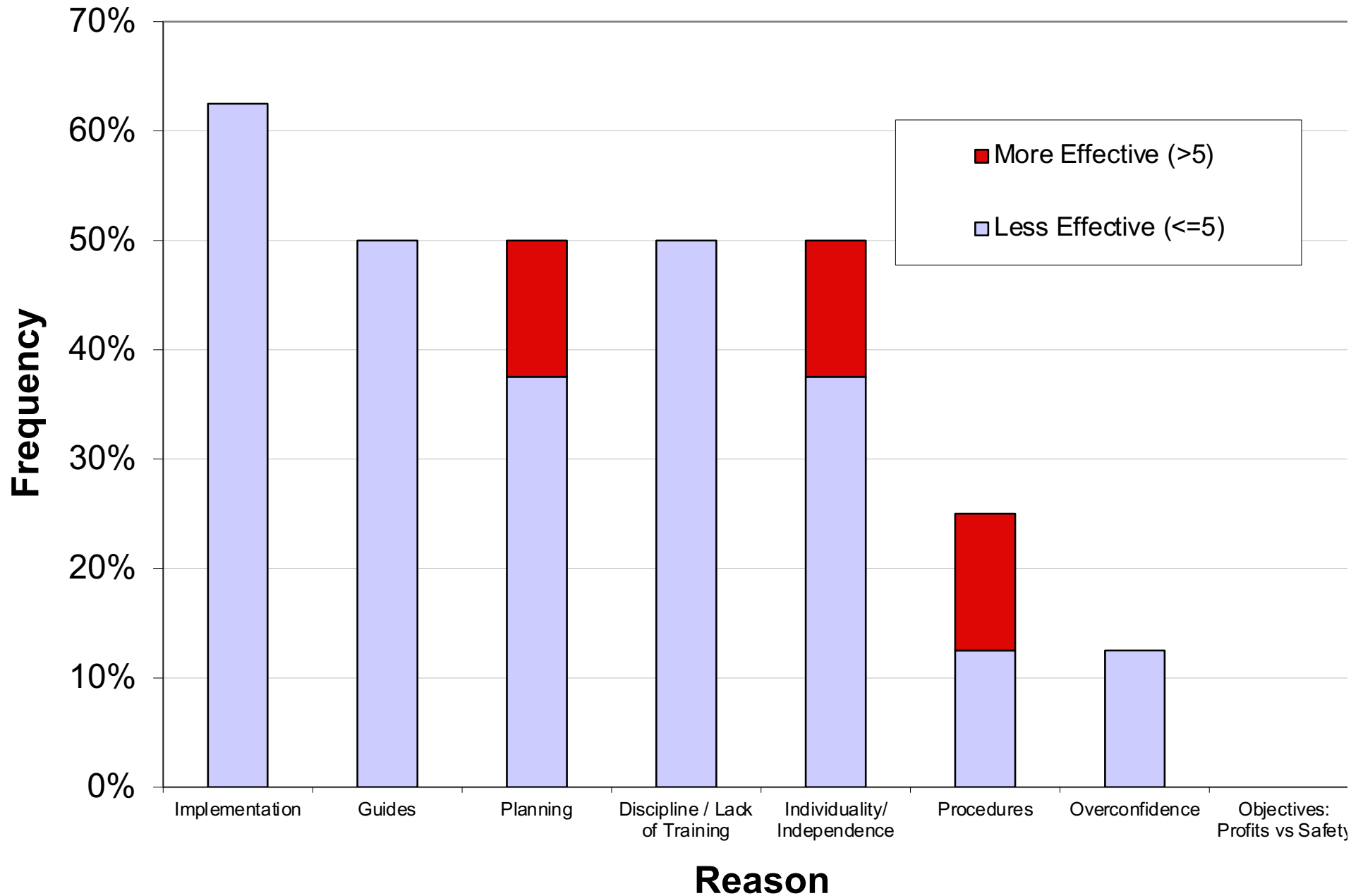
(Fischer = 3.43, Hall = 5.15)



Rob Hall Effectiveness Rating: Reasons



Scott Fischer Effectiveness Rating: Reasons



“8000ers” Statistics

summit	height	ascents	deaths	Rate	Through
Everest	8848	12636			
K2	8611	800			
Kangchenjunga	8586	568			
Lhotse	8516	1197			
Makalu	8463	718			
Cho Oyu	8188	3990			
Dhaulagiri I	8167	691			
Manaslu	8163	650			
Nanga Parbat	8126	400			
Annapurna I	8091	478			
Gasherbrum I	8080	350			
Broad Peak	8047	406			
Gasherbrum II	8035	1000			
Shishapangma	8027	350			

Adventure Consultants Brochure

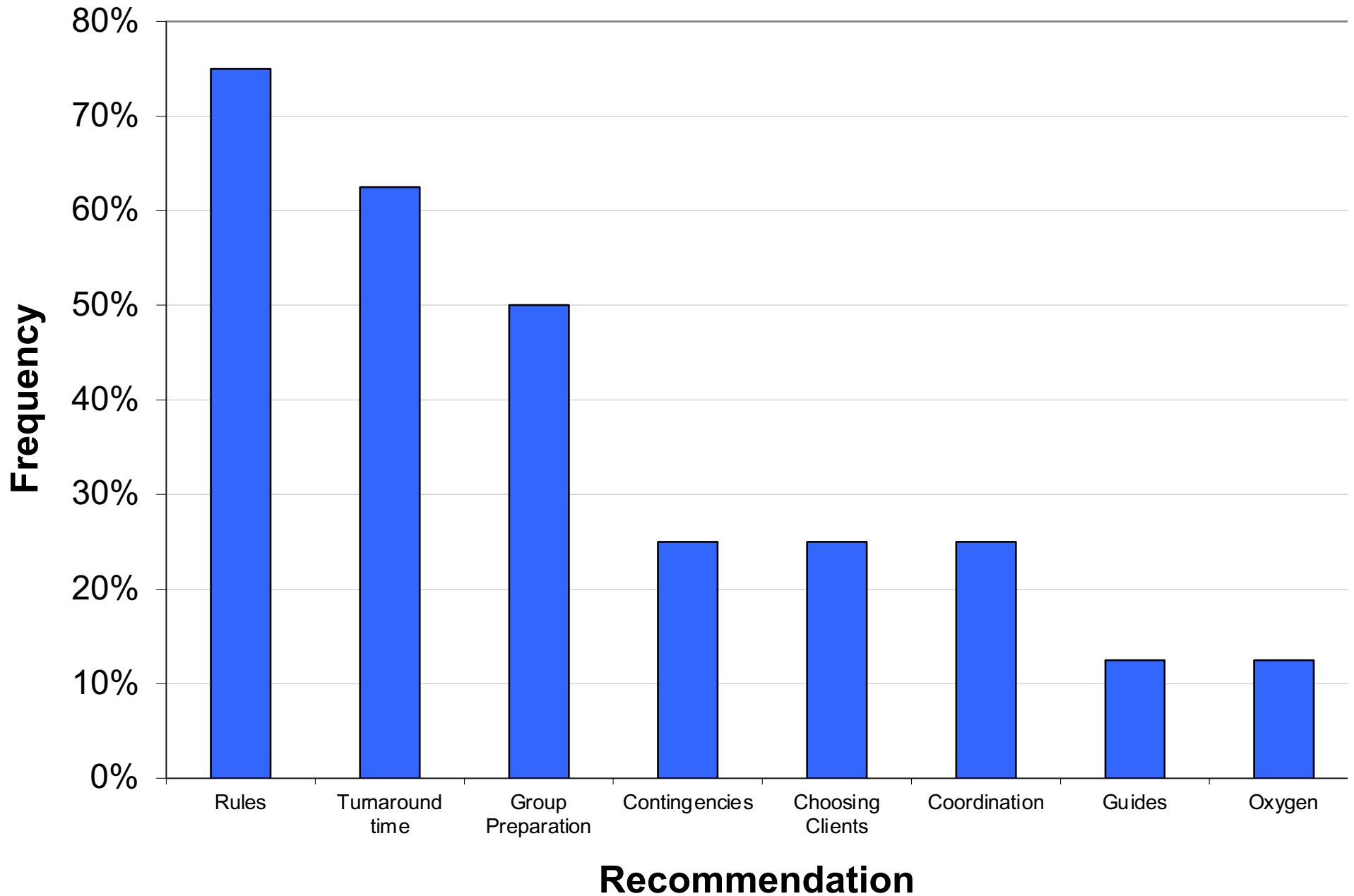
The world leader in Everest climbing with more ascents than any other organisation...

So, you have a thirst for adventure! Perhaps you dream of visiting seven continents or standing on top of a tall mountain. Most of us never dare act on our dreams and scarcely venture to share them or admit to great inner yearnings.

Adventure Consultants specialises in organising and guiding mountain climbing adventures. Skilled in the practicalities of developing dreams into realities, we work with you to reach your goal. We will not drag you up a mountain—you will have to work hard – but we guarantee to maximise the safety and success of your adventure.

For those who dare to face their dreams, the experience offers something beyond the power of words to describe. We invite you to climb your mountain with us.

What would you do differently?



Optimism at Play

In not one of our undertakings did we ever attempt anything unless we were 100 percent certain of achieving success. We never overestimated ourselves as, alas, so many mountaineers often do, frequently with irrevocable consequences.

Peter Habeler,
The Lonely Victory (1978, p.16)



Optimism at Play II



I believe 100 percent I'm coming back... My wife believes 100 percent I'm coming back. She isn't concerned about me at all when I'm guiding because I'm gonna make all the right choices. When accidents happen, I think it's always human error. So that's what I want to eliminate. I've had lots of climbing accidents in my youth. You come up with lots of reasons, but ultimately it's human error.

Hey, experience is overrated. It's not the altitude that's important, it's your attitude, bro. You'll do fine. You've done some pretty sick climbs—stuff harder than Everest. We've got the big E figured out, we've got it totally wired. These days, I'm telling you, we've built a brick road to the summit.

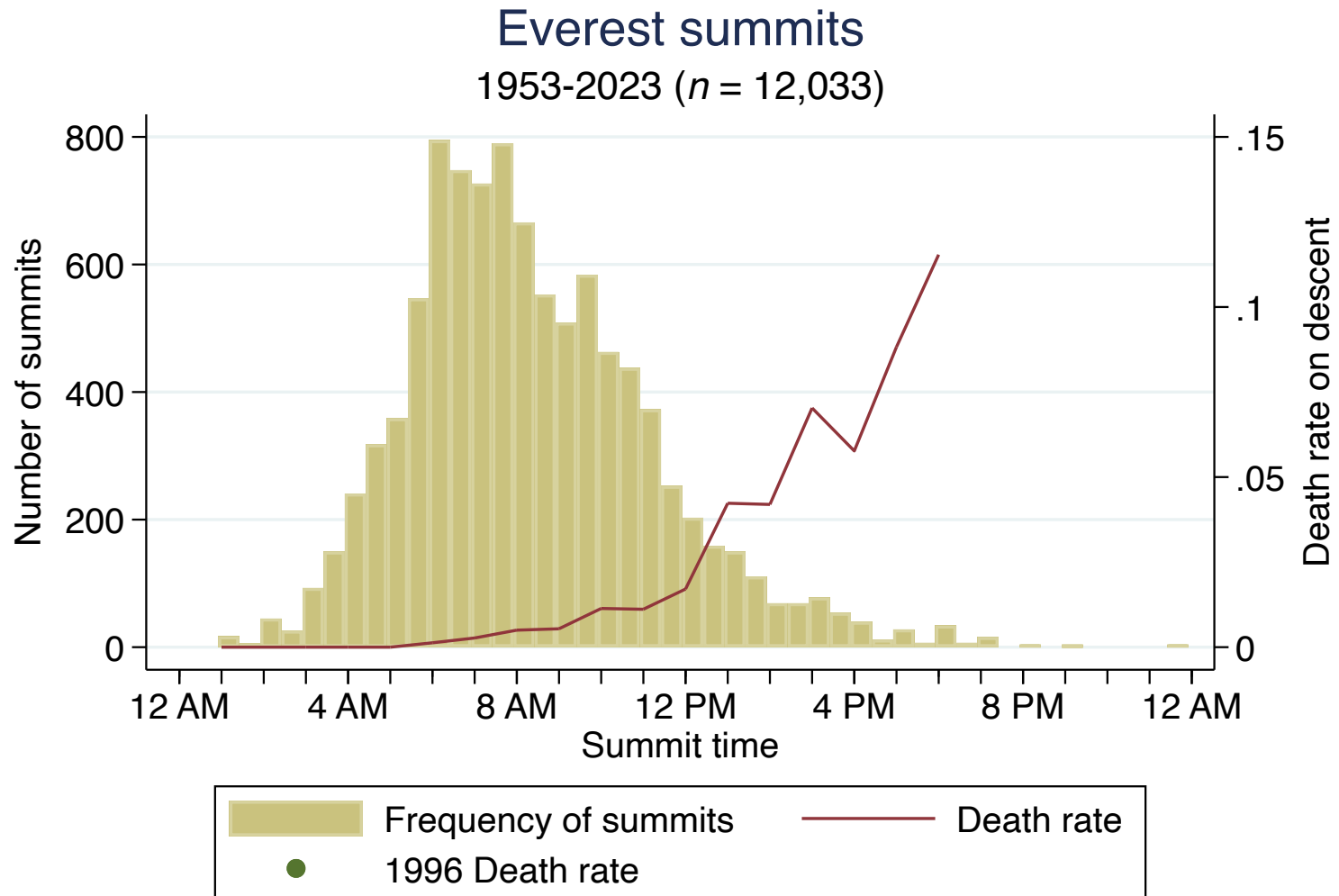
- Scott Fischer, from Seattle, Spring 1996,
Into Thin Air

Decision Making on Everest

At the core of mountaineering is an ongoing process of evaluation. One tries to determine the location of the line between skills one has and the risks one faces. The secret is to recognize the line and to know when to turn back. Too soon, and nothing is gained; too late, and you've been reckless and die.

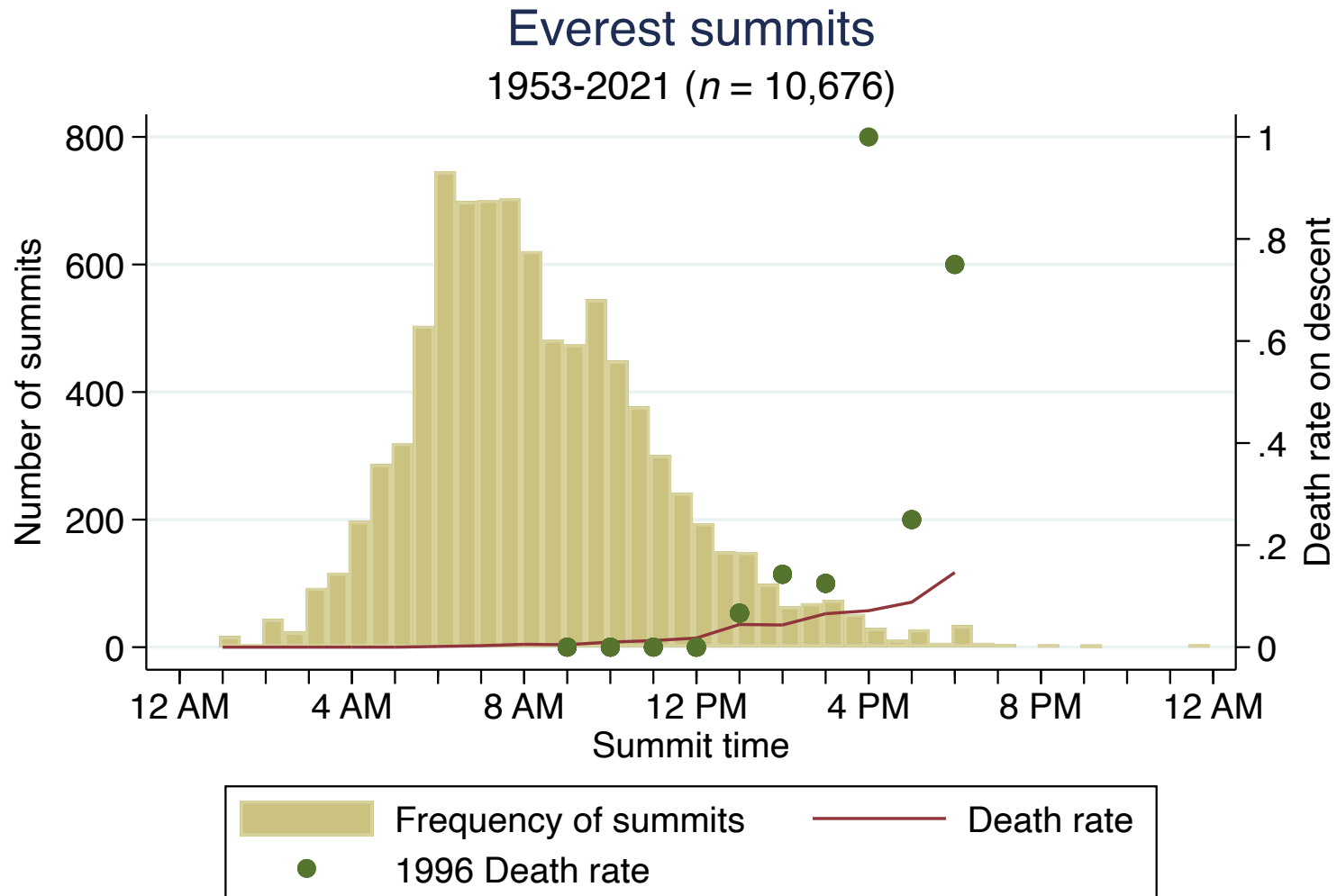
Morrow, cited in Nelson et al, 1990

Data on Turning Around



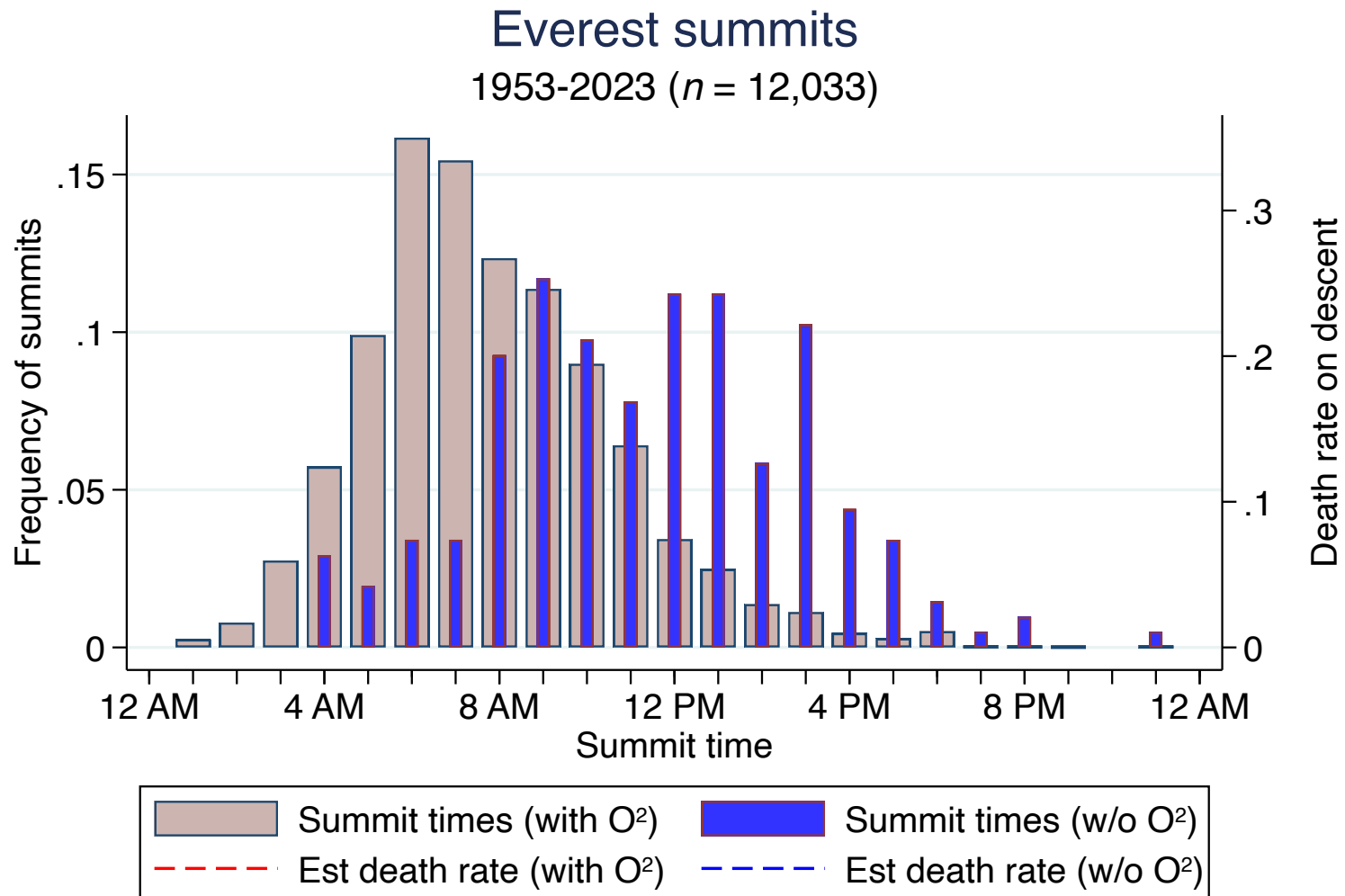
Note: 9,754 of 12,033 summits include summit times

Data on Turning Around



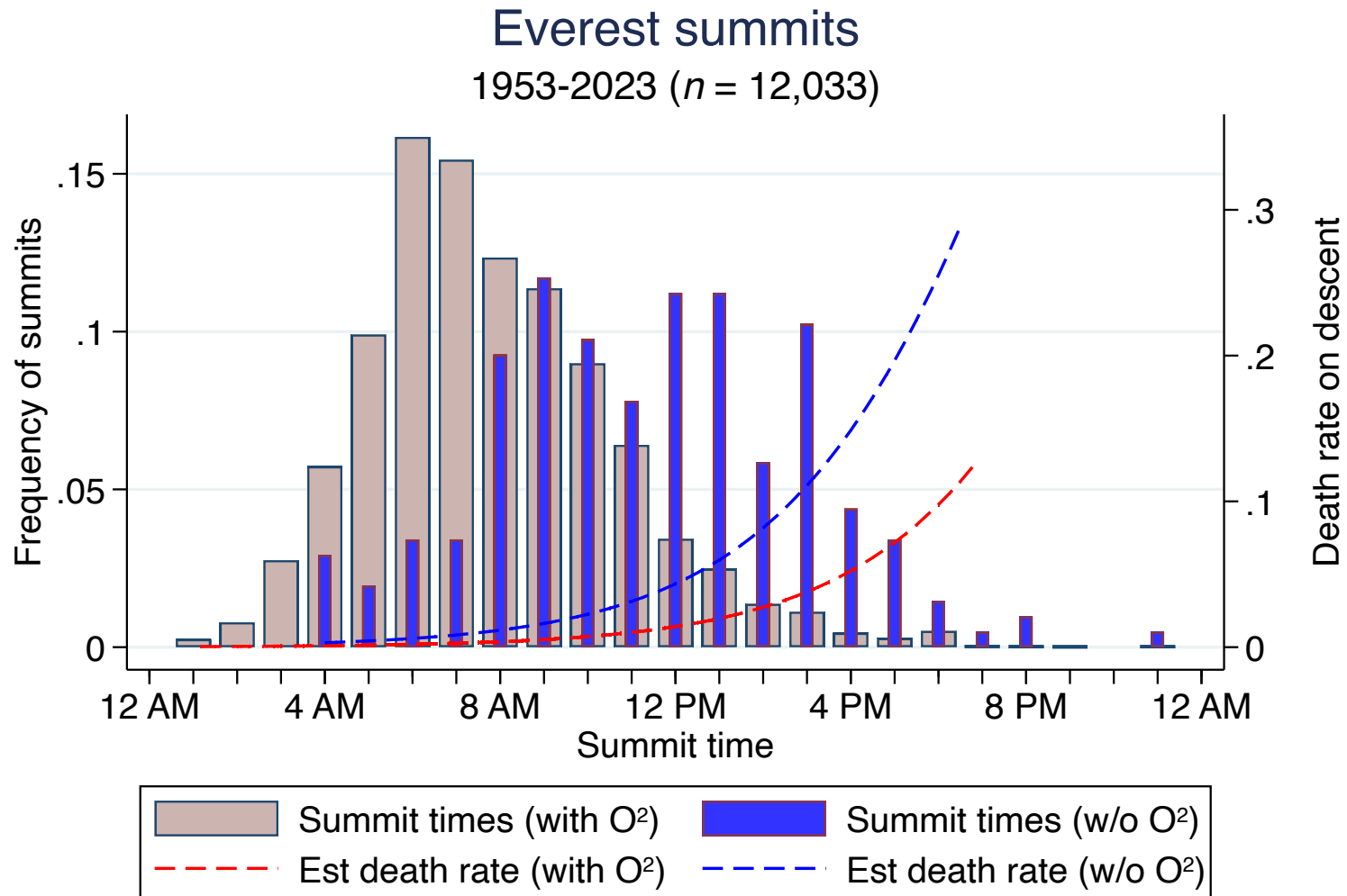
Note: 8,937 of 10,676 summits include summit times

Data on Turning Around



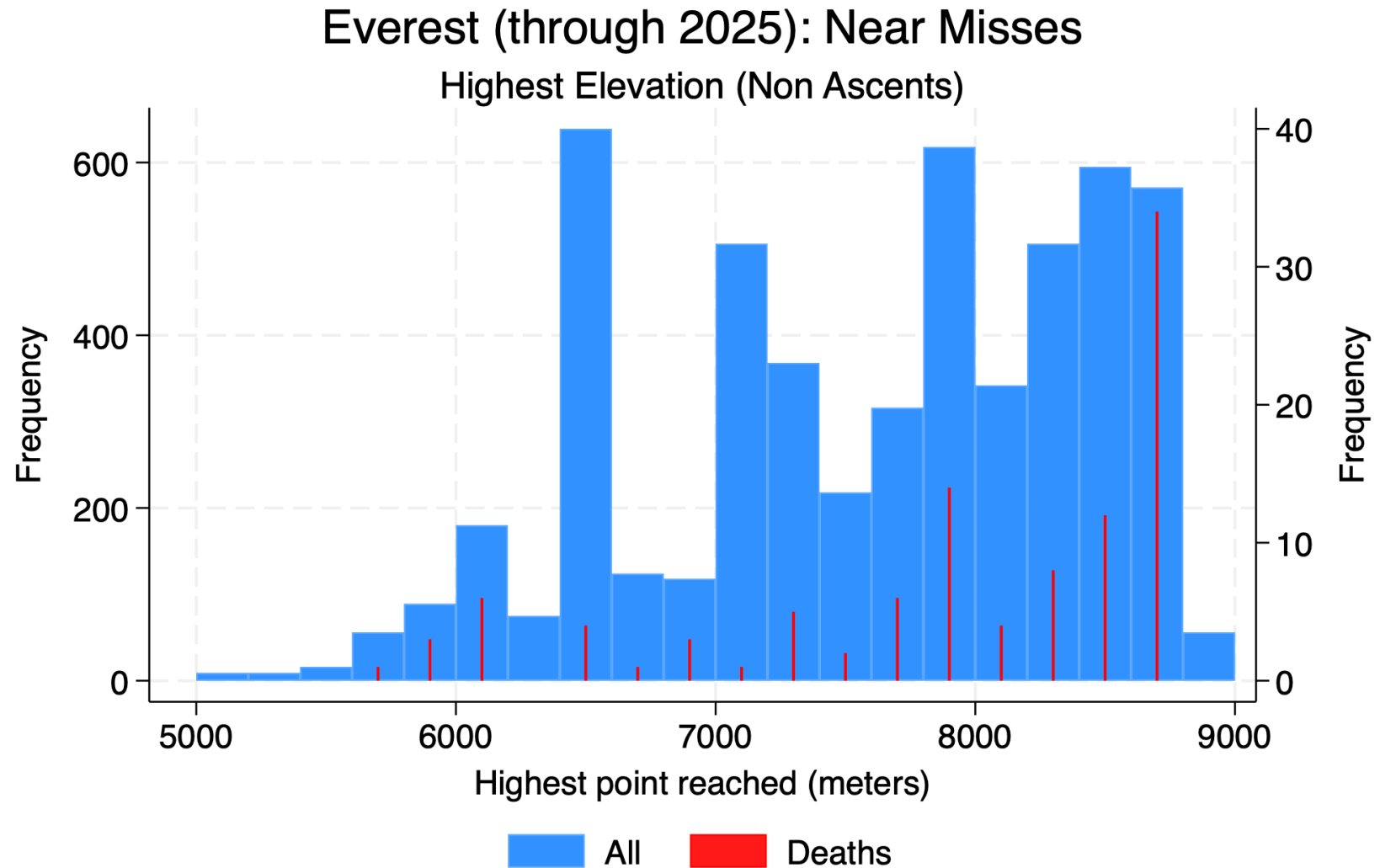
Note: 9,754 of 12,033 summits include summit times

Data on Turning Around

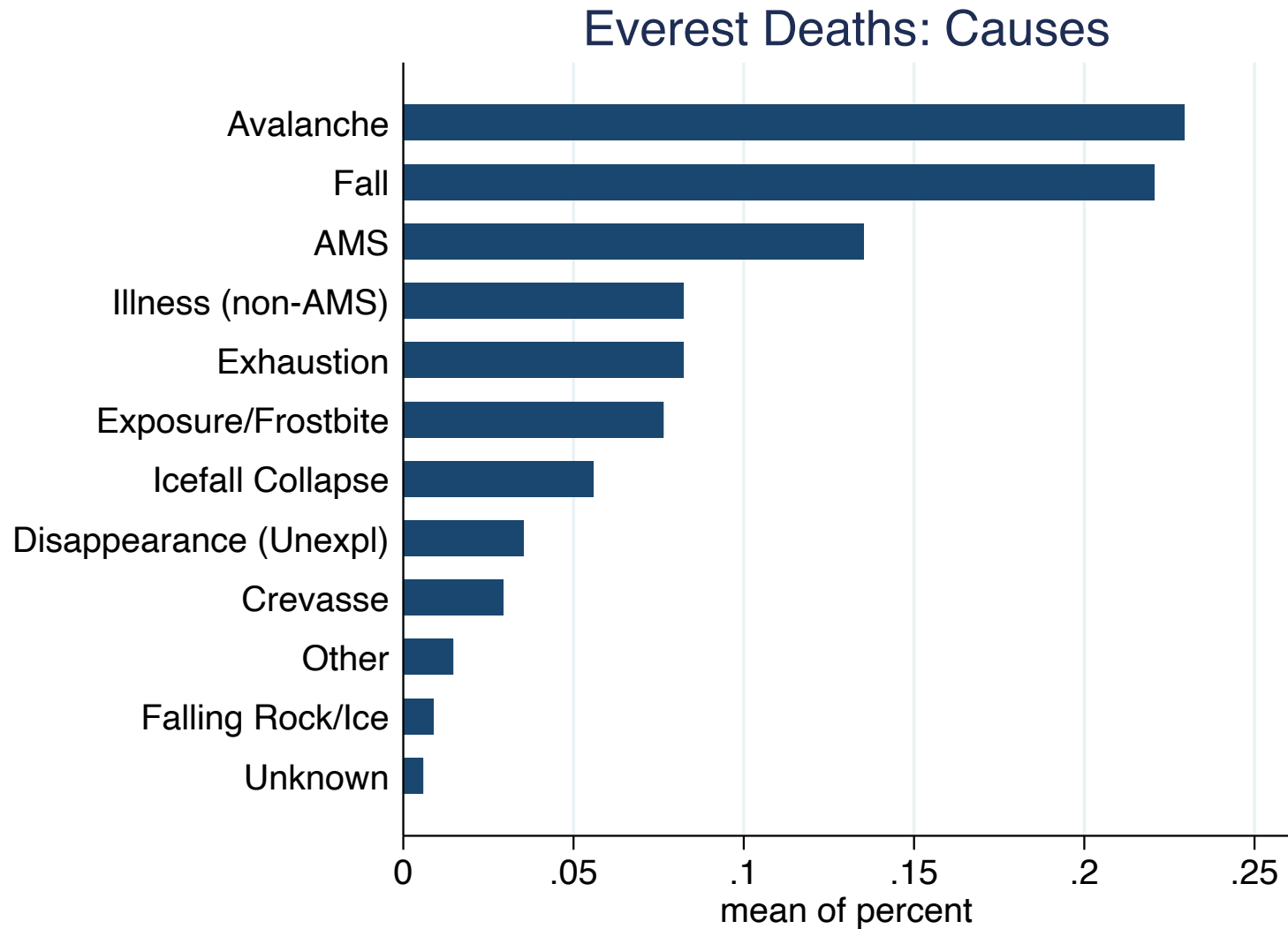


Note: 9,754 of 12,033 summits include summit times

Near Misses



Deaths



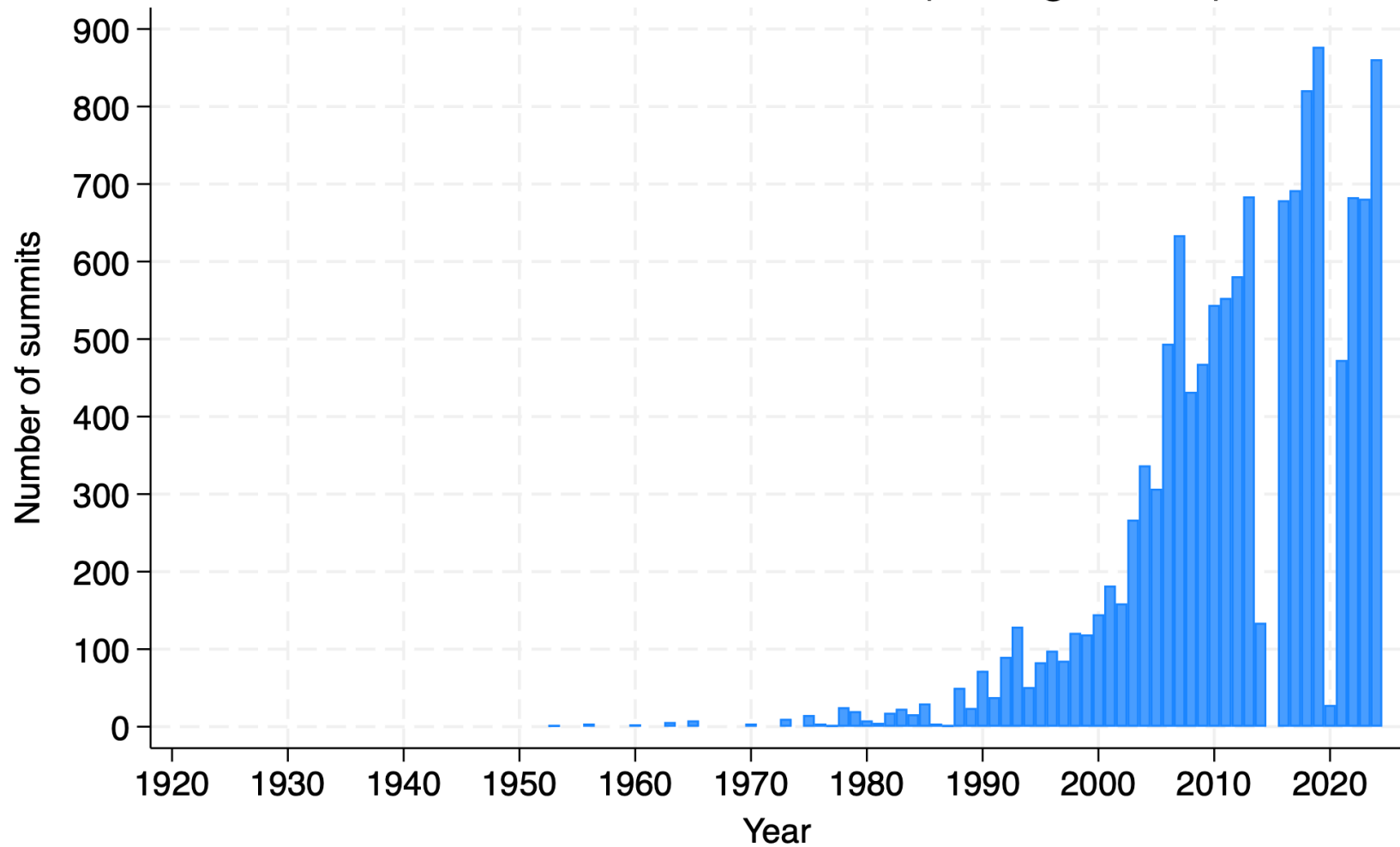
Availability

The first body had left me badly shaken for several hours; the shock of encountering the second wore off almost immediately. Few of the climbers trudging by had given either corpse more than a passing glance. It was as if there were an unspoken agreement on the mountain to pretend that these desiccated remains weren't real—as if none of us dared to acknowledge what was at stake here.”

- Jon Krakauer, *Into Thin Air*

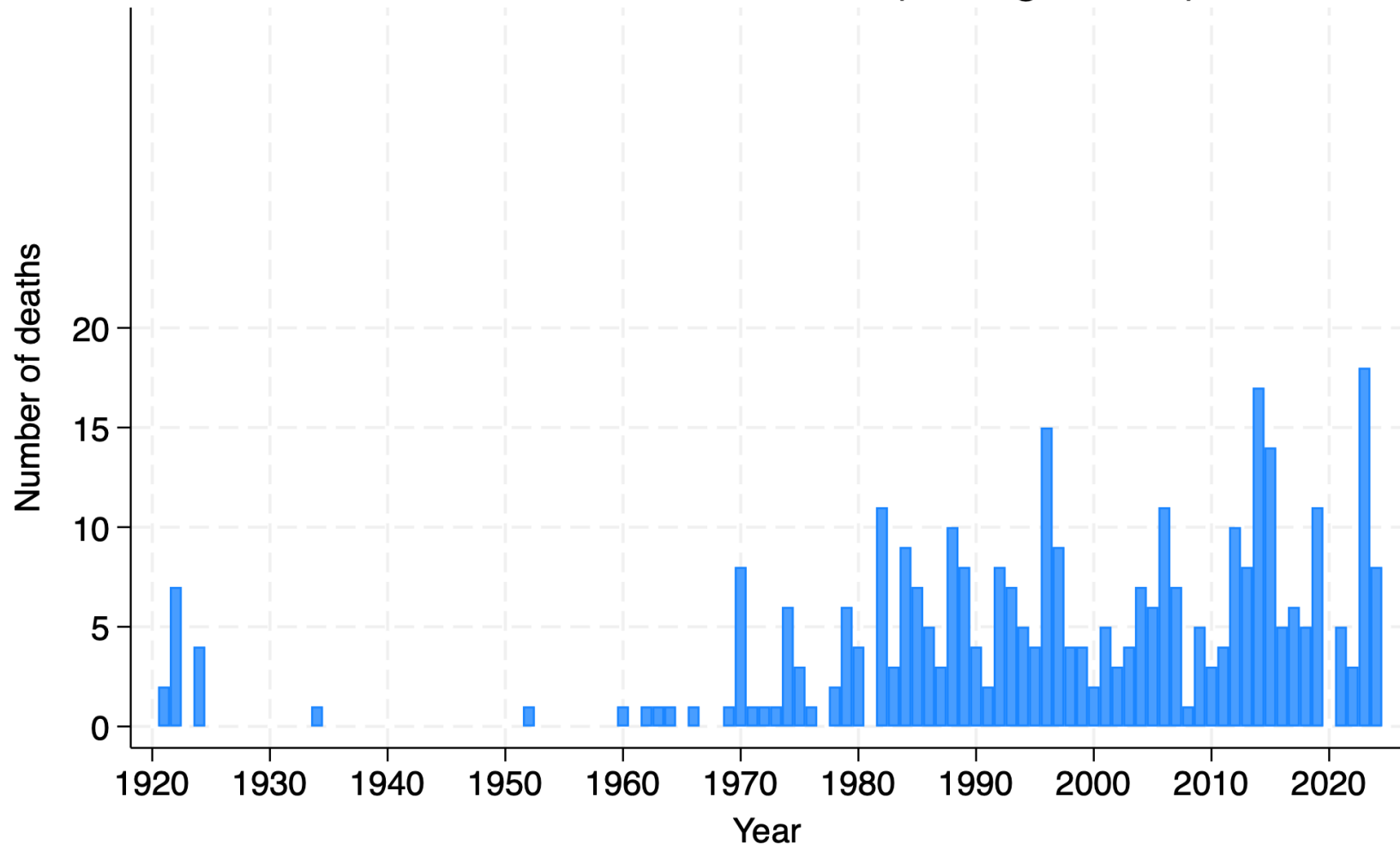
Everest Summits

Everest number of summits (through 2024)



Everest Summits (and Deaths)

Everest number of deaths (through 2024)



Wu's Clues: Part 3



Clue 10 (Going forward): In order to learn, you must learn to ask yourself (and others) why you might be wrong.

Clue 11 (Looking back): In order to learn, you must document and get buy in on your decision making process.

Clue 12: Algorithms aren't perfect, but people are even worse.

Clue 13: Be aware of how other people influence your decisions.
